AI- DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

INDIVIDUAL THESIS

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Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Cinnamon cultivation faces challenges in monitoring plant growth, detecting diseases, and determining the appropriate time for harvesting. Manual assessment can be time-consuming and inconsistent, creating a need for an intelligent system that supports accurate, data-driven decision-making. This research aims to develop a machine learning-based approach for cinnamon growth prediction, disease detection, and robotic harvest guidance, with a focus on assessing bark thickness and plant health. The system development follows a design science and experimental research approach, using environmental parameters, including temperature, humidity, soil moisture, and light intensity, alongside plant images. A Random Forest regression model is employed to predict cinnamon bark thickness, while an EfficientNetB0 deep learning model is used to classify plant images into seven disease-related classes. A classification model is also developed to assess harvest readiness under different decision thresholds. The Random Forest model achieved an R^2 of 0.8393 and a root mean squared error (RMSE) of 0.1648 mm. The initial EfficientNetB0 model achieved 84.07% precision, 86.11% recall, an F1-score of 84.65%, and a receiver operating characteristic area under the curve (ROC-AUC) of 0.9860. The exported disease classification model achieved an F1-score of 78.60% and a ROC-AUC of 0.9824. For harvest-readiness assessment, the binary decision model achieved an F1-score of 96.00%, while a stricter harvest-approval threshold achieved an F1-score of 63.16%. These results indicate the potential of machine learning to support cinnamon growth monitoring, disease identification, and harvest-readiness decisions. The individual contribution is the development and evaluation of these machine learning components to provide decision support for robotic cinnamon harvesting. Although multispectral imaging is included in the research title, its implementation in the reported experiments is not established by the available methodology and results.

Keywords: Cinnamon cultivation, growth prediction, disease detection, deep learning, robotic harvest guidance.

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TABLE OF CONTENTS

DECLARATION AND SUPERVISOR CERTIFICATION i

ABSTRACT ii

ACKNOWLEDGEMENTS iii

TABLE OF CONTENTS iv

LIST OF FIGURES vii

LIST OF TABLES viii

LIST OF ABBREVIATIONS ix

1 INTRODUCTION 1

1.1 Overall Project Context 1

1.2 Individual Component Context 3

1.3 Problem Statement 4

1.4 Research Gap 4

1.5 Aim and Objectives 6

1.6 Research Questions 7

1.7 Scope, Limitations and Assumptions 8

1.8 Individual Contribution 11

2 LITERATURE REVIEW AND RELATED WORK. 13

2.1 Review Method and Scope 13

2.2 Domain and Theoretical Foundations 14

2.3 Related Research 16

2.4 Existing Systems, Products and Techniques 17

2.5 Gap Synthesis and Conceptual Framework 19

3. RESEARCH METHODOLOGY 21

3.1 Research Design 21

3.2 Data, Participants, and Study Materials 22
3.3 Experimental Procedure 24
3.4 Measures and Evaluation Metrics 30
3.5 Data Analysis 32
3.6 Ethics, Privacy, Security, and Responsible AI 33
3.7 Reproducibility and Research Artefacts 33
4 . 4. PRODUCT AND SOLUTION DESIGN 35
4.1 Requirements 35
4.2 Component Architecture within the Complete Solution 37
4.3 Design Alternatives and Decisions 40
4.4 Detailed Design 41
5. IMPLEMENTATION, INTEGRATION AND TESTING 44
5.1 Development Environment and Technology Stack 44
5.2 Component Implementation 46
5.3 Integration with Other Components 48
5.4 Verification and Software Testing 49
5.5 Deployment and Operation 54
5.5.1 Backend Deployment 54
5.5.2 Frontend Deployment 55
5.5.3 Deployment of Supporting Services 55
6 RESULTS AND VALIDATION 56
6.1 Research results 56
6.2 Product and technical results 61
6.3 User or Stakeholder Validation 63
6.4 Integration Results 65
7. DISCUSSION 67

7.1 Interpretation of Findings 67
7.2 Comparison with Literature Baselines and Existing Solutions 67
7.3 Research Contribution 68
7.4 Product Contribution 69
7.5 Limitations and Threats to Validity 69
7.6 Implications 71
8 CONCLUSION AND FUTURE DIRECTIONS 72
8.1 Conclusion 72
8.2 Achievement of objectives 73
8.3 Individual research and product contributions 75
8.4 Future directions 76
REFERENCES 79
GLOSSARY 81
APPENDICES ..... **Error! Bookmark not defined.**
Appendix A - Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance UI Layouts .**Error! Bookmark not defined.**

LIST OF FIGURES

Figure 1 - Overall System Diagram 2
Figure 2 - Individual Component Diagram 3
Figure 3 - Component Architecture Diagram 38
Figure 4 - Scatter Plot of Growth Prediction 57
Figure 5 - Disease classification 59
Figure 6 - Harvest Classification 61

LIST OF TABLES

Table 1 - Comparative Analysis of Techniques 18
Table 2 - Seven Image-Level Classes Of Diseases 22
Table 3 - summarizes the principal technologies 45
Table 4 - Test Cases 50
Table 5 - growth prediction evaluation results 57
Table 6 - disease-classification performance 58
Table 7 - Harvest-classification results 60

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CNN	Convolutional Neural Network
DL	Deep Learning
ESP32	Espressif Systems 32-bit Microcontroller
F1-Score	F1 Score (Harmonic Mean of Precision and Recall)
FN	False Negative
FP	False Positive
IoT	Internet of Things
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
NIR	Near-Infrared
RF	Random Forest
RGB	Red, Green, Blue
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
R^2	Coefficient of Determination
TN	True Negative
TP	True Positive
UAV	Unmanned Aerial Vehicle
UI	User Interface

1 INTRODUCTION

1.1 Overall Project Context

Agriculture plays an important role in economic development, food production, employment, and the sustainability of rural communities. The productivity and quality of agricultural crops depend on several interacting factors, including plant age, environmental conditions, soil characteristics, plant health, and the timing of harvesting. Monitoring these factors is essential for making appropriate cultivation decisions and maintaining consistent crop quality. However, agricultural monitoring can become challenging when it depends primarily on manual observation, individual experience, and periodic physical measurements. These approaches may require considerable time and labor, and the results can vary according to the experience of the person conducting the assessment.

The development of Artificial Intelligence (AI), Machine Learning (ML), computer vision, and Internet of Things (IoT) technologies has created opportunities to improve agricultural monitoring and decision-making. Machine learning can identify patterns within collected data and generate predictions that assist users in interpreting crop conditions. Computer vision can analyze plant images to identify visible characteristics associated with plant health and development. Environmental sensors can provide information such as temperature, humidity, and soil moisture, allowing environmental conditions to be considered alongside plant-related observations. When these technologies are combined, they can provide a more structured approach to agricultural monitoring and decision support.

Smart agriculture involves the use of digital technologies to collect, process, and interpret agricultural information. Rather than depending exclusively on manual observations, a smart agricultural system can combine environmental readings, image-based observations, and predictive models to assist in monitoring crops. For example, environmental information can be used as an input to a growth-estimation model, while image classification can help identify visible symptoms associated with plant diseases. Predictions about crop maturity can also provide information that supports decisions about when harvesting should be considered.

Cinnamon cultivation provides a relevant context for investigating these technologies. Cinnamon is an important agricultural crop, and the quality of the harvested bark is an important consideration in its production. Effective cultivation and harvesting require attention to plant development, bark characteristics, plant health, and the suitability of the plant for harvesting. Assessing these characteristics consistently can be difficult when observations are made manually and at different stages of cultivation. A digital

monitoring system may therefore help organize available information and support more consistent assessments.

The CinnaAI project, titled *Smart Cinnamon Monitoring and Harvest Decision Support*, addresses this context by combining several technology-based functions within a single software prototype. These functions include environmental monitoring, cinnamon bark thickness estimation, image-based disease detection, harvest-readiness assessment, and a robotic harvesting research interface. The system is intended to provide users with information that can assist in understanding plant conditions and making decisions related to cinnamon cultivation and harvesting.

The wider project incorporates a range of technologies, including Python-based machine learning, a web-based user interface, environmental data retrieval, image processing, and application services for recording and presenting prediction results. These technologies support different parts of the monitoring workflow. However, the presence of a digital interface or prediction endpoint does not by itself demonstrate that the underlying agricultural prediction is accurate under field conditions. The models must be evaluated according to the data on which they are trained, the reliability of their input information, and the extent to which their predictions agree with independently observed agricultural outcomes.

Within this wider project, the present dissertation focuses on the machine-learning component responsible for cinnamon growth prediction, disease detection, and harvest guidance. The research considers how data-driven models and image analysis can be used to support these functions, while also examining the limitations that must be addressed before such predictions can be relied upon in practical cinnamon cultivation.

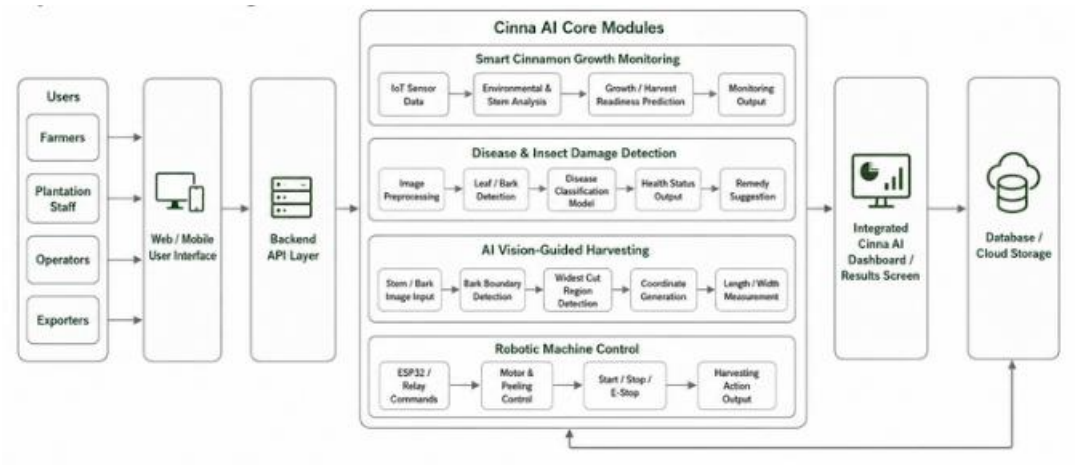


Figure 1 - Overall System Diagram

1.2 Individual Component Context

The individual research component, titled "Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance Using Multispectral Imaging," focuses on three tasks: bark-thickness prediction, image-based disease classification, and harvest-readiness assessment.

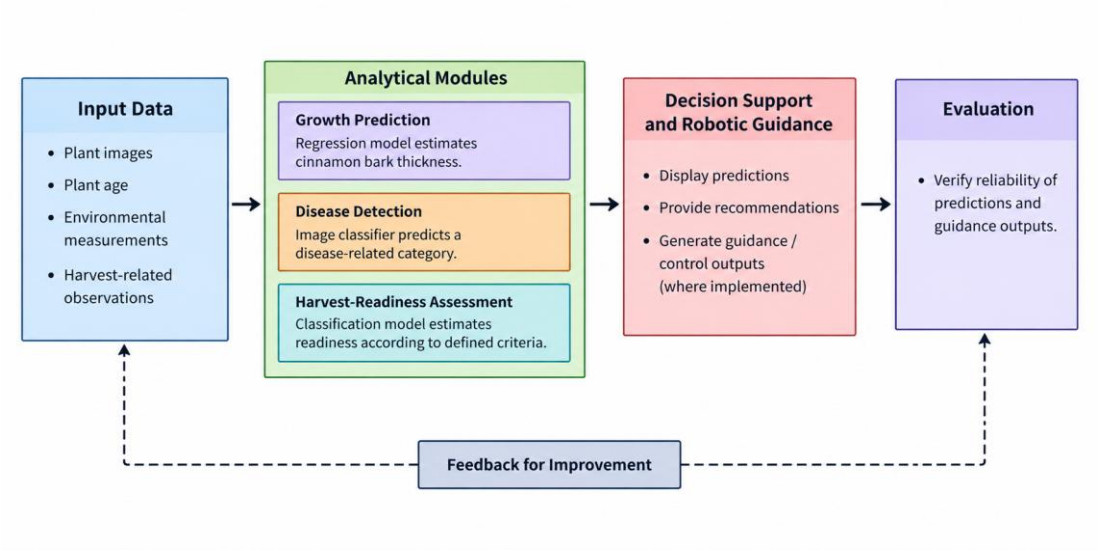


Figure 2 - Individual Component Diagram

The growth-prediction module uses Random Forest regression with inputs such as plant age, temperature, humidity, and soil moisture to estimate bark thickness. However, the model uses generated data based on a predefined mathematical relationship. Therefore, its predictions demonstrate the implemented estimation approach but do not independently establish real-world cinnamon growth accuracy.

The disease-detection module uses an EfficientNetB0-based model to classify uploaded leaf images into seven disease-related classes. Its reliability depends on image quality, dataset representativeness, class distribution, and independent evaluation. Predictions should be treated as model-generated assessments rather than definitive disease diagnoses. Although the research title refers to multispectral imaging, the available implementation description does not confirm the use of multiple spectral bands.

The harvest-readiness module uses Random Forest classification with plant age, rainfall index, phenological stage, and bark browning percentage to generate readiness probabilities and recommendations. Since its training records and labels are generated using predefined conditions, independent field observations and physical peeling assessments are required to establish its practical accuracy.

The component also includes image measurement, cut-depth estimation, and relay-control interaction to investigate robotic harvesting guidance. However, these functions do not establish fully autonomous harvesting or verified field performance. Overall, the component integrates growth estimation, disease classification, and harvest-readiness assessment to support cinnamon cultivation decisions, while distinguishing implemented functionality from capabilities requiring further validation.

1.3 Problem Statement

Cinnamon cultivation requires monitoring plant growth, identifying potential diseases, and determining suitable harvesting times. These tasks involve different inputs, including environmental measurements, plant images, maturity indicators, and physical characteristics. Assessing them separately can make it difficult to combine information into a consistent digital decision-support process.

Growth estimation requires a reliable relationship between plant and environmental variables and independently measured bark thickness. Models trained on generated data may reproduce predefined assumptions without accurately representing real plant growth. Similarly, image-based disease classification can be affected by image quality, lighting, symptom similarity, and dataset limitations.

Harvest-readiness assessment presents another challenge because predefined rules and generated training labels do not necessarily reflect actual peeling readiness. Furthermore, converting predictions into robotic harvesting operations requires accurate measurements, appropriate cutting parameters, reliable control mechanisms, and physical validation.

The central research problem is the need to develop and evaluate an integrated machine-learning-based cinnamon component that supports bark-thickness estimation, disease classification, and harvest-readiness decisions while accounting for data limitations and the gap between software predictions and real-world agricultural performance.

1.4 Research Gap

Machine learning, deep learning, and computer vision offer opportunities for agricultural monitoring. However, the effectiveness of these techniques depends on data quality, valid target variables, suitable evaluation procedures, and practical integration. This research addresses the implementation and evaluation of three related functions within a cinnamon-monitoring prototype.

1.4.1 Bark-Thickness Prediction Validation

The Random Forest regression model estimates bark thickness from plant and environmental inputs using generated data. A key gap is the lack of independent validation against measured cinnamon bark thickness. Field measurements across different plant ages and environmental conditions are required to establish real-world predictive accuracy.

1.4.2 Reliable Disease Classification

The EfficientNetB0 model classifies leaf images into seven disease-related classes. Its reliability requires evaluation using representative images, verified labels, appropriate training and test separation, and class-level performance metrics. If multispectral imaging is intended to improve classification, its contribution must be demonstrated using verified spectral inputs and suitable comparisons.

1.4.3 Harvest-Readiness Validation

The Random Forest classifier generates readiness probabilities and recommendations from plant and environmental indicators. Because its current training labels are based on predefined conditions, independent field assessments and physical peeling outcomes are needed to determine whether its predictions reflect actual harvesting suitability.

1.4.4 Robotic Harvesting Integration

Robotic harvesting involves measurement, target positioning, cutting-depth selection, actuator control, and physical outcome verification. Although the prototype includes image-measurement, numerical estimation, and relay-control functions, complete autonomous harvesting has not been established by the available implementation evidence. Further testing is required to evaluate guidance accuracy, control reliability, and harvesting outcomes.

1.4.5 Multispectral Imaging Verification

Multispectral imaging captures information from multiple wavelength bands beyond standard RGB channels. However, the current implementation description confirms image-based classification without establishing the use of multiple spectral bands. The actual sensor inputs and processing pipeline must therefore be verified before claiming multispectral functionality or benefits.

1.4.6 Overall Research Gap

The overall research gap is the need to implement and evaluate an integrated cinnamon-monitoring component combining bark-thickness prediction, disease classification, and harvest-readiness assessment. The study examines how these functions operate together while distinguishing performance on generated data from validated agricultural outcomes. Its contribution is limited to the implemented features and available evidence, with real-world prediction accuracy, multispectral benefits, and autonomous harvesting requiring further validation.

1.5 Aim and Objectives

1.5.1 Research Aim

The primary aim of this research is:

To develop and evaluate a machine-learning-based cinnamon monitoring and decision-support component that estimates bark thickness, classifies disease-related plant images, and assesses harvest readiness to support robotic harvesting guidance, while identifying the limitations of the available data and validation procedures.

1.5.2 Research Objectives

The research is guided by the following objectives:

1. **Investigate cinnamon growth-estimation requirements** by identifying relevant input variables, including plant age, temperature, humidity, and soil moisture, and examining their use in the implemented bark-thickness estimation workflow.
2. **Develop and examine a bark-thickness estimation model** using Random Forest regression, documenting the training-data construction, target definition, prediction outputs, and available evaluation measures.
3. **Implement image-based disease classification** using the EfficientNetB0 model to classify uploaded plant images into the seven disease-related classes supported by the current implementation.
4. **Examine disease-classification performance** using appropriate evaluation measures and distinguish the results associated with model development from the behaviour of the exported inference model.
5. **Develop a harvest-readiness assessment workflow** using the selected plant and environmental indicators to generate a readiness probability and decision-support output.

6. **Examine the limitations of the harvest-readiness model** by considering the origin of its training labels, the use of predefined conditions, and the need for independent validation against actual harvesting outcomes.
7. **Investigate the role of prediction outputs in robotic harvest guidance** by documenting the available image-measurement, cut-depth estimation, and control-interface functions, while distinguishing implemented software capabilities from unverified physical harvesting operations.
8. **Evaluate the component within its verified scope** by reporting the available model and software results accurately, identifying data-related limitations, and defining the additional evidence required to establish practical agricultural performance.
9. **Identify the status of multispectral imaging in the implementation** by determining whether the image inputs include multiple spectral bands and, where this is not established, documenting the limitation rather than attributing unverified benefits to multispectral data.

1.6 Research Questions

The research questions are formulated to correspond with the three main functions of the individual component and the validation issues identified above.

RQ1: How can plant age and environmental information be used by a Random Forest regression model to estimate cinnamon bark thickness within the implemented system?

RQ2: How effectively does the EfficientNetB0-based image-classification model classify uploaded cinnamon plant images into the seven supported disease-related classes?

RQ3: How does the performance of the initial disease-classification model compare with that of the exported inference model, based on the available evaluation measures?

RQ4: How can plant age, rainfall index, phenological stage, and bark browning percentage be used to generate harvest-readiness predictions within the implemented decision-support component?

RQ5: To what extent do the available training and evaluation procedures establish the reliability of the growth-estimation and harvest-readiness models for real cinnamon cultivation conditions?

RQ6: How can growth estimates, disease-classification outputs, and harvest-readiness recommendations support a robotic harvesting workflow, and what additional validation is required before autonomous harvesting can be claimed?

RQ7: Does the current implementation use multispectral image information, and what evidence would be required to establish whether multispectral data improve disease classification or other plant-monitoring outcomes?

These questions are primarily exploratory and evaluative. They do not assume that every component has already demonstrated reliable field performance. Instead, they guide the examination of implemented functionality, model results, data limitations, and requirements for further validation.

1.7 Scope, Limitations and Assumptions

1.7.1 Scope of Research

This research focuses on the machine-learning and decision-support functions of the CinnaAI cinnamon-monitoring prototype. The individual component covers three principal areas: growth estimation, disease-related image classification, and harvest-readiness assessment with robotic harvesting guidance.

The growth-estimation scope includes the use of plant age, temperature, humidity, and soil moisture as model inputs; the implementation of Random Forest regression; and the generation of bark-thickness and related growth outputs. The evaluation considers the model as implemented and records the distinction between performance on generated targets and accuracy against independently measured cinnamon growth.

The disease-detection scope includes the EfficientNetB0-based classification workflow, uploaded plant images, seven disease-related classes, prediction outputs, and the evaluation information available for the initial and exported models. The study considers the importance of image quality, class definitions, data representativeness, and the difference between a predicted disease-related class and a confirmed diagnosis.

The harvest-readiness scope includes the implemented classifier using plant age, rainfall index, phenological stage, and bark browning percentage. It also considers the readiness probability, status and action messages, and separate maturity-related guidance provided through the interface. The analysis distinguishes model-generated outputs from independently verified peeling-readiness results.

The robotic harvesting guidance scope includes the software functions relevant to image measurement, numerical cut-depth estimation, and relay-control interaction.

These are considered in relation to the broader objective of linking agricultural assessment with harvesting support. The research does not assume that these functions demonstrate a fully autonomous, physically validated harvesting robot.

Multispectral imaging is considered in relation to the stated research title and the available implementation evidence. Unless the actual input data and processing pipeline confirm the use of multiple spectral bands, the disease-classification experiments are described as image-based classification without asserting that multispectral data were used.

1.7.2 Limitations of the Research

The main limitations of the research are as follows:

- **Generated growth-model data:** The available implementation report indicates that the bark-thickness model is trained using generated samples and a predefined mathematical relationship. Performance against these targets does not establish predictive accuracy for measured cinnamon growth in the field.
- **Generated harvest-readiness labels:** The current harvest classifier uses generated records and labels constructed from predetermined conditions. These labels do not represent independently observed physical peeling-test outcomes.
- **Disease dataset and evaluation evidence:** The reliability of disease classification depends on image provenance, class balance, annotation quality, and the separation of training and evaluation samples. The available evidence must be reported according to the evaluation procedure actually performed.
- **Differences between model versions:** The initial disease model and the exported inference model are distinct evaluation subjects. Their results should not be combined or treated as interchangeable without evidence that the same evaluation data and procedure were used.
- **Unconfirmed multispectral input:** The available implementation description does not establish the use of multiple spectral bands in the reported image-classification experiments. Therefore, the specific contribution of multispectral imaging cannot be claimed without further implementation evidence.
- **Limited field validation:** The available project report does not establish that all three machine-learning functions have been validated using a representative, independently collected field dataset. Real-world generalization remains an important area for further investigation.
- **Robotic validation:** The existence of image-measurement, cut-depth estimation, and relay-control software does not establish the accuracy, safety, or success rate of a complete physical harvesting operation.

- **Environmental variability:** Cinnamon plants may be affected by differences in growing conditions, soil characteristics, cultivation practices, seasonal conditions, and measurement procedures. Models developed within limited data conditions may not generalize to other settings.
- **Input reliability:** Incorrect, missing, outdated, or inconsistently measured input values may affect the resulting predictions. The reliability of environmental readings and manually entered harvest indicators is therefore relevant to interpretation.
- **Decision-support limitations:** A disease-classification result or harvest-readiness score is a model output and should not automatically be treated as a definitive agricultural decision. Independent inspection and appropriate professional judgement may still be necessary.

1.7.3 Research Assumptions

The research adopts the following assumptions within the limits of the current implementation:

1. The input variables selected by each model are available in the format required by the corresponding prediction workflow.
2. The generated training data and labels are suitable for testing the software implementation and the model behaviour within their defined synthetic domain, but are not assumed to represent all real cinnamon-growing conditions.
3. The seven disease-related classes are treated according to the labels supported by the implemented model. Their meaning and correctness depend on the underlying dataset and labelling process.
4. Harvest-readiness outputs are interpreted as recommendations generated from the model and its defined input conditions, rather than as independently confirmed peeling outcomes.
5. Environmental readings are assumed to represent the values supplied to the system at the time of prediction. Their calibration, freshness, and representativeness must be verified before making stronger claims about field accuracy.
6. Robotic guidance functions are assumed to be decision-support or prototype-control functions unless their connection to a physical mechanism and the resulting harvesting performance have been independently demonstrated.
7. Multispectral imaging is not assumed to have been implemented in the reported experiments unless the image-acquisition hardware, spectral channels, and data-processing workflow provide evidence of its use.

These assumptions establish a clear boundary between the functionality that can be examined in the current prototype and the additional evidence needed to establish agricultural and robotic performance.

1.8 Individual Contribution

The overall CinnaAI project combines several functions for cinnamon monitoring, environmental observation, disease classification, harvest assessment, and robotic harvesting research. The individual contribution of this dissertation is focused on the machine-learning and decision-support aspects associated with growth estimation, disease detection, and harvest guidance.

1.8.1 Growth-Estimation Component

The growth-estimation contribution concerns the use of Random Forest regression to generate bark-thickness estimates from the selected plant and environmental inputs. It includes documenting the input variables, training-data construction, prediction workflow, and available performance measures. It also identifies the limitations of evaluating a model against generated targets and the need for independent bark-thickness measurements to establish real-world predictive accuracy.

1.8.2 Disease-Detection Component

The disease-detection contribution concerns the implementation and examination of an EfficientNetB0-based model for classifying plant images into seven disease-related classes. It includes describing the image input and prediction workflow, examining the available evaluation results, and distinguishing the initial model from the exported inference model. It also considers the importance of class labels, image representativeness, evaluation design, and the limitations of treating an image prediction as a confirmed diagnosis.

1.8.3 Harvest-Readiness Assessment

The harvest-readiness contribution concerns the use of a Random Forest classifier to estimate readiness from plant age, rainfall index, phenological stage, and bark browning percentage. The component generates a probability-based readiness output and corresponding status or action information. The research documents the model's operation and examines how its generated training labels affect the interpretation of its results.

1.8.4 Robotic Harvest Guidance

The contribution to robotic harvesting focuses on examining how model outputs and plant measurements may inform a harvesting workflow. This includes documenting the relationship between readiness assessment, image measurement, numerical cut-depth estimation, and the available control interface. The contribution is limited to the functions supported by the implementation evidence and does not claim that a complete autonomous harvesting process has been validated.

1.8.5 Evaluation and Critical Analysis

An important part of the individual contribution is the critical interpretation of model performance. The research distinguishes generated-data performance from results obtained using independent real-world observations. It also considers differences between the initial and exported disease models, the validity of harvest-readiness labels, and the evidence needed to assess practical generalization.

This evaluation-oriented contribution is important because a model can produce outputs and achieve a performance score without necessarily demonstrating the same level of accuracy in practical agricultural conditions. By documenting the available evidence and its limitations, the dissertation provides a clearer basis for understanding what the prototype currently achieves and what requires further investigation.

1.8.6 Contribution to the Wider CinnaAI Project

The individual component contributes to the broader CinnaAI objective of bringing agricultural monitoring and harvesting decision support into a digital system. Growth estimates, disease-related image predictions, and harvest-readiness outputs provide complementary information that can help users examine plant conditions. The robotic harvesting interface represents a possible connection between this information and physical harvesting operations.

The principal contribution is therefore the implementation and evaluation of these machine-learning-based functions within the defined project scope, together with an explicit assessment of their data requirements, limitations, and validation needs.

2 LITERATURE REVIEW AND RELATED WORK.

2.1 Review Method and Scope

This literature review examines research related to machine learning in agriculture, cinnamon disease detection, plant growth prediction, multispectral imaging, and computer-vision-based robotic harvesting. These research areas were selected because they correspond to the principal functions of the proposed component: bark-thickness prediction, disease classification, harvest-readiness assessment, and robotic harvest guidance.

The review uses the twenty references supplied for this research as its initial literature set. The publications are organized into five broad categories: cinnamon disease detection and plant classification [1]–[3], plant growth prediction and intelligent sensing [4]–[7], multispectral imaging for plant disease detection [8]–[13], robotic harvesting and computer vision [14]–[19], and EfficientNet-based plant disease classification [20].

The literature is examined according to the problem addressed, the techniques used, the relevance of the findings to agricultural monitoring, and the limitations that may affect application to cinnamon cultivation. Particular attention is given to the relationship between input data, model selection, prediction outputs, and evaluation methods. The review also distinguishes research conducted specifically on cinnamon from studies involving other crops. This distinction is necessary because methods that perform well on one crop may require modification and additional validation before being applied to another.

The review considers both conventional machine-learning approaches and deep-learning techniques. Conventional machine learning is relevant to predicting numerical values from structured agricultural measurements, while deep learning is relevant to extracting features from plant images. Research on multispectral imaging is included to examine the potential benefits of spectral information for plant-health monitoring. Robotic harvesting literature is reviewed to identify the technical requirements for translating agricultural predictions into harvesting guidance and physical actions.

The scope is limited to methods that can contribute to the three principal functions of the proposed component. Research on unrelated agricultural applications is considered only when it provides a useful methodological foundation. For example, plant-growth prediction research involving lettuce or groundnut can inform the selection of predictive techniques, but it does not directly validate a cinnamon bark-thickness model [5], [7]. Similarly, fruit-harvesting studies provide relevant principles for

computer vision and robotic guidance, but their harvesting procedures cannot automatically be transferred to cinnamon [14]–[19].

2.2 Domain and Theoretical Foundations

2.2.1 Machine Learning in Precision Agriculture

Machine learning enables computational models to learn patterns from data and make predictions or classifications. In agriculture, it supports plant monitoring, growth estimation, disease detection, and management decisions. Regression predicts continuous values, such as bark thickness, while classification assigns categories, such as disease type or harvest readiness.

The proposed cinnamon component applies both approaches. Previous studies have investigated machine learning for agricultural sensing, growth prediction, and environmental analysis [4]–[7]. However, model performance depends on data quality, representativeness, and validation. Therefore, each model must be evaluated using suitable metrics and data relevant to its intended application.

2.2.2 Cinnamon Growth and Bark-Thickness Prediction

Bark-thickness prediction can be formulated as a regression problem using plant and environmental characteristics, including plant age, temperature, humidity, and soil moisture. Random Forest regression is a potential approach because it combines multiple decision trees to model nonlinear relationships between input features and numerical outputs.

Previous studies have demonstrated machine-learning approaches to plant-growth prediction and agricultural sensing [4]–[7]. However, predicting biomass or vegetative growth is different from predicting cinnamon bark thickness. The proposed model therefore requires cinnamon-specific measurements and appropriate evaluation.

Predicted thickness should be compared with measured values using regression metrics such as mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination ((R^2)), where appropriate. The training-data source must also be documented because results from generated data do not independently establish real-world prediction accuracy.

2.2.3 Deep Learning for Cinnamon Disease Detection

Computer-vision-based disease detection analyses plant images to identify patterns associated with different conditions. Traditional methods extract features such as

colour and texture, whereas convolutional neural networks (CNNs) learn image features during training.

Thandapani et al. investigated texture-based feature extraction and conventional classifiers for cinnamon disease classification [1]. Bhat and Sujithra examined multiple CNN architectures for cinnamon bark disease classification [2]. These studies provide a foundation for exploring image-based disease detection. The proposed component uses EfficientNetB0, a CNN architecture that balances model capacity and computational efficiency.

Classification performance can be evaluated using precision, recall, F1-score, and ROC-AUC, where appropriate. These metrics should be interpreted alongside class distribution, dataset quality, and the evaluation procedure. Lighting, background variation, image resolution, and symptom similarity may affect performance, so independent testing is necessary to assess generalization.

2.2.4 Multispectral Imaging for Plant Disease Detection

Multispectral imaging captures information across selected wavelength bands, potentially providing additional information beyond the red, green, and blue channels of conventional cameras. Previous studies have investigated multispectral techniques for plant-virus detection, disease diagnosis, coffee leaf rust, and wheat yellow rust [8]–[11]. Further research has explored close-range spectral imaging and hyperspectral remote sensing for precision agriculture [12], [13].

The effectiveness of these techniques depends on sensor characteristics, spectral-band selection, calibration, preprocessing, and reliable reference labels. For the proposed cinnamon component, the imaging method must be described according to the actual implementation. If the classifier uses only RGB photographs, it should be identified as an RGB-based model. Multispectral performance can only be claimed if spectral data were acquired and used in the model pipeline.

2.2.5 Harvest Readiness and Robotic Guidance

Harvest-readiness assessment uses selected plant characteristics to determine whether harvesting conditions are satisfied. Relevant features may include plant age, development stage, visible indicators, and environmental measurements. A classification model can estimate readiness categories, but labels generated from predefined rules may only demonstrate that the model reproduces those rules. Independent agricultural observations are needed to establish practical validity.

Robotic harvesting extends this process through target recognition, localization, position estimation, planning, and physical control. Research on fruit-picking robots and harvesting systems highlights the importance of integrating perception, manipulation, and detachment [14], [15]. Further studies examine fruit recognition, grasp estimation, and picking-pose estimation [16]–[18], while Ghonimy and Abdel-Baky discuss the integration of perception, manipulation, and detachment [19].

For the proposed cinnamon component, harvest-readiness prediction must therefore be distinguished from robotic guidance and physical harvesting. Each function requires appropriate evaluation to establish its accuracy and reliability.

2.3 Related Research

2.3.1 Cinnamon Disease Detection and Plant Classification

Thandapani et al. explored feature extraction and machine-learning classifiers for cinnamon disease classification [1]. Bhat and Sujithra investigated CNN architectures and data augmentation for cinnamon bark disease classification [2]. Yang et al. applied deep convolutional neural networks to distinguish visually similar *Cinnamomum* species [3]. These studies demonstrate the potential of image processing and deep learning for cinnamon analysis. The proposed component builds on this research by using EfficientNetB0 for disease classification. The additional EfficientNet-related reference [20] should be verified before citing specific findings.

2.3.2 Plant Growth Prediction and Intelligent Sensing

Nesteruk et al. investigated plant-growth prediction using intelligent embedded sensing [4]. Liang et al. explored machine-learning methods for predicting lettuce biomass across different climate zones [5]. Fujita et al. examined growth-environment prediction [6], while Bala et al. investigated groundnut growth prediction using IoT and machine learning [7]. These studies support data-driven agricultural prediction. However, their findings cannot be directly transferred to cinnamon bark-thickness estimation, which requires appropriate cinnamon-specific measurements and validation.

2.3.3 Multispectral Imaging and Plant Disease Detection

Peng et al. investigated multispectral imaging for early plant-virus detection [8]. Zhang et al. studied multispectral diagnosis of *Panax notoginseng* blight [9], while Da Silva Soares et al. examined coffee leaf rust detection [10]. Nguyen et al. investigated wheat yellow rust using multispectral UAV imagery [11]. Scutelnic et al. explored close-range multispectral imaging [12], and Terentev et al. reviewed hyperspectral remote

sensing for early disease detection [13]. These studies highlight the potential of spectral information in plant-health assessment. However, their applicability to cinnamon must be validated using suitable sensors and disease-specific data.

2.3.4 Robotic Harvesting and Computer Vision

Tang et al. reviewed visual recognition and localization methods for fruit-picking robots [14]. Droukas et al. surveyed robotic harvesting technologies [15], while Kang et al. investigated real-time fruit recognition and grasping estimation [16]. Alaaudeen et al. examined intelligent robotic harvesting and grasping prediction [17]. Zhang et al. reviewed vision-based apple-harvesting systems [18], and Ghonimy and Abdel-Baky discussed the integration of perception, manipulation, and detachment [19]. These studies provide a foundation for understanding how computer vision and robotic control can support automated harvesting.

2.4 Existing Systems, Products and Techniques

2.4.1 Conventional Machine Learning and Deep Learning

Conventional machine-learning methods, such as Random Forest, are suitable for structured agricultural data and regression tasks. CNN-based methods learn image features for classification. Cinnamon disease studies have explored both feature-based classification and deep-learning approaches [1], [2], while Yang et al. demonstrated deep learning for *Cinnamomum* species identification [3]. In the proposed component, Random Forest is used for bark-thickness prediction, whereas EfficientNetB0 supports disease classification. Each model requires task-appropriate evaluation metrics.

2.4.2 Sensor-Based Growth Prediction

Sensor-based prediction uses measurements such as temperature, humidity, and soil moisture to estimate plant-growth characteristics. Previous studies have investigated embedded sensing, environmental prediction, and IoT-based growth modelling [4], [6], [7]. For cinnamon bark-thickness prediction, model reliability depends on the quality of the input data and actual thickness measurements. Simulated data, if used, must be clearly identified, with conclusions limited accordingly.

2.4.3 Image-Based Disease Classification

Image-based disease detection uses handcrafted features, traditional classifiers, or deep-learning models. Cinnamon-specific research supports the application of feature extraction and CNN architectures to disease classification [1], [2]. EfficientNet-related research provides additional context for deep-learning approaches to plant disease

detection [20]. The proposed EfficientNetB0 model should be evaluated using clearly defined disease classes, independent test data, and suitable classification metrics.

2.4.4 Multispectral and Hyperspectral Imaging

Multispectral and hyperspectral imaging can provide additional information for detecting plant diseases. Previous studies have applied these techniques to plant-virus detection, blight, coffee leaf rust, and wheat yellow rust [8]–[11]. Further research has explored precision-agriculture imaging and hyperspectral disease detection [12], [13]. However, these approaches may require specialized sensors and calibration. If the proposed model uses only RGB images, its results should not be described as multispectral performance.

2.4.5 Harvest-Readiness and Robotic Harvesting

Harvest-readiness assessment can use predefined rules or machine-learning models to determine whether a plant is ready for harvesting. Robotic harvesting additionally involves target recognition, localization, pose estimation, motion planning, and detachment [14]–[19]. These functions require separate evaluations. Therefore, the proposed cinnamon component should assess harvest-readiness prediction independently from cutting-position estimation, cut-depth calculation, and connected-mechanism control.

2.4.6 Comparative Analysis of Techniques

Table 1 - Comparative Analysis of Techniques

Technique	Main input	Main output	Key consideration
Random Forest regression	Plant and environmental variables	Estimated bark thickness	Requires reliable target measurements and regression evaluation
Feature extraction with machine learning	Extracted image features	Predicted image class	Depends on feature quality and classifier suitability
CNN / EfficientNetB0	Labelled plant images	Disease-class prediction	Requires representative images and appropriate validation

Multispectral analysis	Images from selected spectral bands	Spectral or spatial–spectral features	Requires verified multispectral acquisition
Harvest-readiness classification	Plant and harvest-related variables	Readiness category	Requires valid labels and meaningful decision criteria
Vision-based robotic guidance	Images and target-related information	Target location or guidance output	Requires appropriate localization and guidance evaluation
Integrated robotic harvesting	Perception, planning, and control outputs	Physical harvesting action	Requires end-to-end system testing

The comparison indicates that no single technique addresses every requirement of the proposed component. Regression, disease classification, harvest-readiness assessment, and robotic guidance serve different purposes. Their integration can support a common decision-support workflow, but the reliability of the overall system depends on the performance and validation of each stage.

2.5 Gap Synthesis and Conceptual Framework

2.5.1 Synthesis of Research Gaps

The reviewed literature establishes the potential of machine learning, computer vision, spectral sensing, and robotics in agriculture. However, several gaps remain for the proposed cinnamon component.

Gap 1: Cinnamon Bark-Thickness Prediction. Existing studies demonstrate machine-learning approaches to agricultural growth prediction [4]–[7], but their applicability to cinnamon bark-thickness estimation requires validation using cinnamon-specific measurements.

Gap 2: Robust Disease Classification. Previous research has explored feature extraction and deep learning for cinnamon disease classification [1], [2], as well as visual identification of *Cinnamomum* species [3]. The proposed classifier requires evaluation using representative images, balanced classes, and independent test data.

Gap 3: Multispectral Imaging Verification. Studies demonstrate the potential of multispectral and hyperspectral techniques for plant-disease detection [8]–[13].

However, their implementation in the proposed component must be confirmed through the actual sensors, spectral channels, and input data.

Gap 4: Harvest-Readiness Validation. Growth-prediction research provides relevant modelling approaches [4]–[7], but harvest readiness requires clearly defined criteria and reliable reference labels. The model must be tested against independently established readiness measurements.

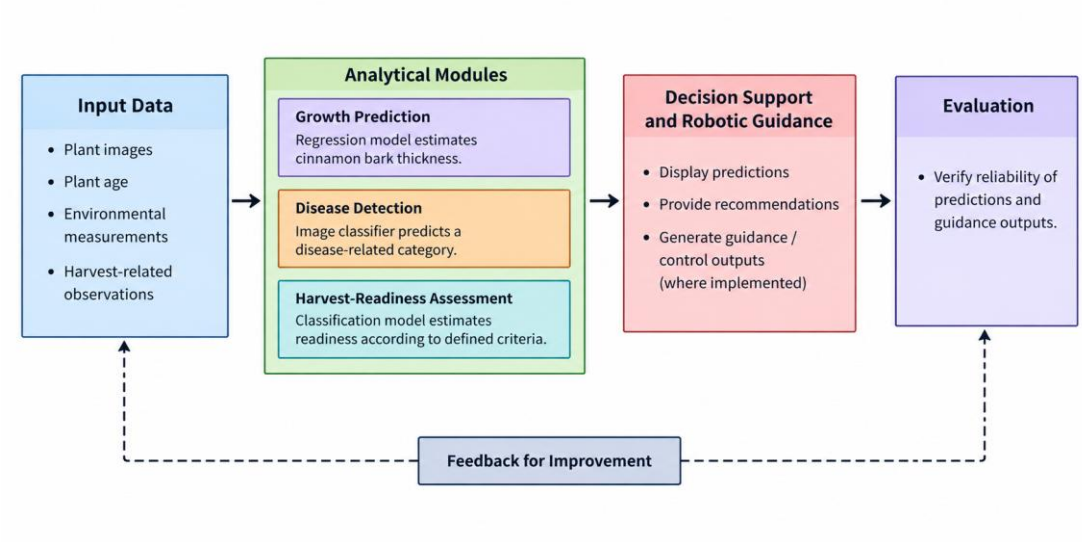
Gap 5: Robotic Guidance Integration. Robotic harvesting research covers recognition, localization, planning, and detachment [14]–[19]. A readiness prediction alone cannot guarantee accurate cutting or successful harvesting; each guidance and control function requires separate evaluation.

Gap 6: Integrated System Evaluation. Existing studies address different tasks and datasets. The proposed component therefore requires consistent module integration and separate evaluations of prediction performance, disease classification, interface functionality, guidance accuracy, and physical harvesting performance.

These gaps motivate an integrated cinnamon component combining bark-thickness prediction, disease classification, and harvest-readiness assessment, with guidance and control capabilities evaluated according to the functions actually implemented.

2.5.2 Conceptual Framework

The proposed conceptual framework connects agricultural data acquisition with three analytical modules and a decision-support stage. Evaluation is included to establish



3. RESEARCH METHODOLOGY

This chapter describes the research design, study materials, experimental procedures, mathematical models, evaluation methods, ethical considerations, and reproducibility measures used to develop CinnaAI. The system combines multispectral image processing, cinnamon leaf disease and condition classification, bark-thickness prediction, harvest-readiness assessment, and robotic harvest guidance.

The methodology distinguishes data preparation, model development, application integration, and system evaluation. It also differentiates successful software operation from demonstrated predictive accuracy and real-world agricultural validity.

3.1 Research Design

This research adopts an applied machine learning and experimental development approach to implement and evaluate an integrated cinnamon monitoring and harvest decision-support system.

The system includes three main analytical functions: image-based disease and condition classification, bark-thickness regression, and harvest-readiness classification. Their outputs are presented through the CinnaAI application and may support robotic guidance where the relevant interfaces are implemented.

The research activities include:

1. Acquiring and preparing agricultural images and numerical data.
2. Preprocessing inputs for image classification and numerical prediction.
3. Developing or integrating machine learning models.
4. Generating predictions, confidence indicators, and recommendations.
5. Integrating the models with the application and available robotic interfaces.
6. Evaluating model performance and software functionality.
7. Examining data quality, uncertainty, and limitations.

3.1.1 Research Approach

The study combines quantitative model evaluation with implementation-based testing. Disease classification is assessed using classification metrics, bark-thickness prediction using regression metrics, and harvest readiness using classification measures. Software testing evaluates data processing, API responses, and application integration.

Each reported result must be associated with the relevant model artifact, dataset partition, and evaluation procedure. Successful software execution alone does not establish biological accuracy or safe robotic operation.

3.1.2 Overall System Workflow

The system follows six stages: data acquisition, input validation and preprocessing, model inference, decision-support processing, robotic guidance where implemented, and evaluation. This workflow connects the individual analytical components while allowing each model to be assessed independently.

3.2 Data, Participants, and Study Materials

3.2.1 Study Materials

The study uses cinnamon leaf images, multispectral data where supported, plant and environmental measurements, bark-thickness reference values, harvest-readiness labels, machine learning models, and software components.

The main software resources include TensorFlow/Keras, Scikit-learn, Python, FastAPI, and the CinnaAI web interface. Supporting data services and robotic communication interfaces are included where implemented.

Each component requires documented input features, data sources, target definitions, preprocessing methods, and evaluation procedures. The study primarily uses plant-related data and does not primarily involve human participants.

3.2.2 Disease and Condition Classification

The retained EfficientNetB0 model supports seven image-level classes:

Table 2 - Seven Image-Level Classes Of Diseases

Class label	Description
`healthy_cinnamon`	Healthy cinnamon leaf
`leaf_blight`	Leaf-blight-like condition
`leaf_miner_attack`	Visible leaf-miner damage
`leaf_patches_fungal`	Fungal patch- or spot-like condition
`lower_leaf_gall`	Gall-like structures on the lower leaf surface
`non_cinnamon`	Unsupported or non-cinnamon material
`upper_leaf_gall`	Gall-like structures on the upper leaf surface

These classes define the model's supported outputs and should not automatically be interpreted as laboratory-confirmed diagnoses. The classifier does not directly perform lesion segmentation, bounding-box detection, lesion counting, or independent disease-severity measurement.

The final dataset description should report actual image counts, class distributions, image sources, labeling procedures, and acquisition conditions.

3.2.3 Multispectral Image Data

Multispectral imaging captures information from selected spectral bands that may reveal plant characteristics not readily distinguishable in conventional images. The study documents the acquisition device, spectral bands, calibration, band alignment, transformations, and input representation according to the implemented workflow.

The retained EfficientNetB0 model accepts a (224\times224\times3) image. Therefore, its direct input is a three-channel representation, and the model architecture alone does not demonstrate that it processes every acquired spectral band. Any transformation from multispectral data to this representation must be documented. Performance improvements attributed to multispectral information require comparative experiments.

3.2.4 Growth-Prediction Data

The growth-prediction component estimates cinnamon bark thickness from the features supplied to its regression model. The dataset description must specify predictor variables, units, record counts, bark-thickness reference sources, preprocessing, and training, validation, and test partitions.

The provenance of the target values is particularly important. Field-measured thickness values must be distinguished from generated or rule-derived targets, as performance on generated targets does not establish accuracy against independent field measurements.

3.2.5 Harvest-Readiness Data

The harvest-readiness component predicts a readiness category using the features available in the implemented model. The dataset description must identify these features, class distribution, preprocessing, and label-generation method.

Labels may originate from field observations, expert assessments, or predefined rules. If rule-derived labels are used, evaluation demonstrates how well the model reproduces those rules rather than independently proving real-world harvesting readiness.

3.2.6 Software and Model Resources

The disease-classification artifact is `backend/cinnamon_multi_part_model.h5`, implemented using TensorFlow/Keras and based on EfficientNetB0. It accepts three-channel images of size (224\times224) and produces seven Softmax outputs.

The backend validates and preprocesses uploaded images before inference, while the frontend presents the results. The wider application includes numerical prediction, harvest-readiness, data-handling, and robotic guidance components according to their implemented interfaces.

3.3 Experimental Procedure

The procedure covers data preparation, model development, inference, application integration, and evaluation.

3.3.1 Data Acquisition and Preparation

Leaf images are collected or received and organized according to the supported classification labels. Where multispectral data are used, acquisition settings, calibration, and band transformations must be recorded.

Images are checked for corruption, unreadable content, and invalid formats. Numerical datasets are examined for missing values, duplicate records, inconsistent units, and invalid observations. All exclusions and preprocessing decisions should be documented.

3.3.2 Image Preprocessing

The retained disease-model inference pipeline performs the following operations:

1. Validates the uploaded image and supported format.
2. Applies file-size limits and rejects empty or invalid inputs.
3. Decodes the image and corrects orientation using available EXIF metadata.
4. Converts the image to RGB.
5. Resizes it to (224\times224) pixels using nearest-neighbour resampling.
6. Converts pixel values to a `float32` array.
7. Adds a batch dimension.

The retained inference pipeline passes pixel values in the range 0–255 without dividing by 255. This must remain consistent with the evaluated model's training configuration. Any multispectral-to-three-channel transformation must also be documented and reproduced consistently.

3.3.3 Dataset Partitioning and Quality Control

The documented disease-model pipeline specifies an approximate class-wise split of 70% training, 15% validation, and 15% testing, using random seed 42. Training data optimize model parameters, validation data support model selection, and test data evaluate the selected model.

Image deduplication uses hashes of normalized pixel data and dimensions. Exact same-class duplicates are skipped, while cross-class duplicate conflicts are recorded. However, this method may not detect near-duplicates or photographs of the same plant. Where possible, images from the same plant, site, or acquisition session should be grouped to reduce data leakage.

The final report should confirm the actual split used for the reported metrics and provide class counts and exclusion details. Similar leakage checks should be applied to growth and harvest-readiness datasets at the plant or observation level.

3.3.4 Model Development and Mathematical Formulation

3.3.4.1 Disease and Condition Classification

The disease classifier uses EfficientNetB0 as a convolutional feature extractor, followed by global average pooling, batch normalization, dropout with a rate of 0.35, and a seven-unit Softmax output.

The documented training configuration uses ImageNet-pretrained weights, Adam optimization, a learning rate of 0.001, sparse categorical cross-entropy, a batch size of 32, and up to 20 initial training epochs. Class weighting and early stopping are used, with checkpoint selection based on validation loss.

The recorded fine-tuning stage uses a learning rate of (10^{-5}), MixUp with (\operatorname{Beta}(0.4,0.4)), label smoothing of 0.05, and up to 15 additional epochs. Selected later backbone layers are trainable while backbone batch-normalization layers remain frozen. Checkpoint selection uses validation accuracy. These settings should be attributed to the evaluated artifact only when supported by the relevant training records.

3.3.4.2 Global Average Pooling

Global Average Pooling (GAP) converts the spatial feature maps into a compact feature vector by calculating the average activation of each channel:

$$z_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F_{ijc}.$$

where (F_{ijc}) represents the activation at spatial position ((i,j)) in channel (c), and (H) and (W) denote the height and width of the feature map, respectively. The documented pooled representation contains 1,280 features.

3.3.4.3 Softmax Classification

The model converts the output logits into normalized scores using the Softmax function:

$$p_k = \frac{\exp(a_k)}{\sum_{j=1}^{7}\exp(a_j)}, \qquad \hat{y} = \underset{k}{\operatorname{argmax}} p_k.$$

where (a_k) is the logit for class (k), and (p_k) is its corresponding Softmax score. The predicted class is selected as follows:

$$m = p_{(1)} - p_{(2)}, \qquad \text{review} = (p_{(1)} < 0.70) \vee (m < 0.15).$$

The highest Softmax score serves as a confidence indicator. However, it should not automatically be interpreted as a calibrated probability that the prediction is correct.

3.3.4.4 Cross-Entropy Loss

Cross-entropy measures the difference between the target labels and the model's predicted class scores:

$$L_{CE} = -\sum_{k=1}^{7} y_k \log p_k.$$

where (y_k) is the target value and (p_k) is the predicted score for class (k). Sparse categorical cross-entropy is documented for initial training, while categorical cross-entropy is used during fine-tuning.

3.3.4.5 Class Weighting

Inverse-frequency class weighting is used to address differences in the number of training samples across classes:

w_k=\frac{N}{Kn_k},\qquad K=7\tag{3.6}

where (N) represents the total number of training images and (n_k) represents the number of images in class (k). This approach assigns greater weight to underrepresented classes. The actual class counts should be used when reporting the calculated weights.

3.3.4.6 MixUp and Label Smoothing

MixUp is applied during fine-tuning to combine pairs of training images and their corresponding labels. The mixing coefficient is sampled from a Beta distribution:

$$L_{CE} = -\sum_{k=1}^{7} y_k \log p_k.$$

The initial integer-label loss is the equivalent sparse form. Initial class weights are:

$$w_k = \frac{N}{Kn_k}, \quad K = 7,$$

These techniques help regularize training and reduce overfitting. However, they do not introduce additional independently collected field observations.

3.3.4.7 Growth Prediction

The bark-thickness regression model estimates thickness using the implemented input features:

where $\boldsymbol{x}$ represents the predictor features and θ denotes the learned model parameters. The prediction function is given by:

$$\hat{t} = f(\boldsymbol{x}; \boldsymbol{\theta})$$

Prediction performance is evaluated using Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |t_i - \hat{t}_i|$$

Root Mean Squared Error (RMSE) is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (t_i - \hat{t}_i)^2}$$

The coefficient of determination (R^2) is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (t_i - \hat{t}_i)^2}{\sum_{i=1}^{n} (t_i - \bar{t})^2}$$

where (t_i) is the reference bark thickness, (\hat{t}_i) is the predicted thickness, (\bar{t}) is the mean reference thickness, and (n) is the number of evaluated observations. These metrics must be interpreted according to the origin of the target values. Performance on generated targets alone does not establish predictive accuracy under real field conditions.

3.3.4.8 Harvest-Readiness Assessment

The harvest-readiness model estimates a readiness category from the implemented input features:

hat{r}=g(\mathbf{u};\phi) \tag{3.15}

where (\mathbf{u}) represents the readiness features and (\phi) denotes the learned model parameters. The resulting category is interpreted according to the criteria defined by the system. Any probability threshold and its effect on precision and recall should be reported. If the training labels are derived from predefined rules, the model requires independent validation against field observations or appropriate expert-defined criteria.

3.3.5 Model Inference and Confidence Assessment

The disease-classification model produces seven Softmax scores, corresponding to the supported image classes. Let (p_{(1)}) and (p_{(2)}) denote the highest and second-

highest scores, respectively. Their difference is calculated as:

$$m=p_{(1)}-p_{(2)} \tag{3.16}$$

The application flags a prediction for review if the highest score is below 0.70 or the difference between the two highest scores is below 0.15:

The system retains the top prediction while indicating when further review is recommended. The *non_cinnamon* class is treated as unsupported input, *healthy_cinnamon* indicates a healthy result, and the remaining classes correspond to their respective condition categories. Explanatory messages and recommendations may accompany the predictions, but they do not constitute independent measurements of disease severity. The thresholds are operational settings and should not be interpreted as clinically or biologically validated standards.

Growth and harvest-readiness predictions are processed through their respective model interfaces. Robotic guidance should be generated only through implemented and tested control logic. A model prediction or guidance recommendation alone does not establish that a physical harvesting action has been completed accurately or safely.

3.3.6 Application Integration

The disease-classification workflow integrates the front-end application with a FastAPI backend. The front end accepts a leaf image and displays the result, while the backend validates the request, preprocesses the image, performs model inference, and returns the classification response.

The growth-prediction and harvest-readiness components follow a similar request-and-response process. Where robotic interfaces are implemented, the mapping between model predictions, guidance outputs, and control commands should be documented separately. Sending a command does not confirm that the physical actuator executed it correctly or that the harvesting operation was safe.

Optional external vision-verification and image-retrieval functions should be described as active system capabilities only when their execution and evaluation can be demonstrated.

3.3.7 Optional Image Retrieval Using FAISS

The project includes source code for optional image retrieval using FAISS, specifically `faiss.IndexFlatIP`, which performs exact inner-product searches over stored feature embeddings. Embeddings can be normalized using the L2 norm:

$$\bar{z} = \frac{z}{\| z \|_2}, \quad s(\bar{z}, \bar{v}) = \bar{z}^T \bar{v} = \frac{z^T v}{\| z \|_2 \| v \|_2}.$$

The documented configuration uses a default similarity threshold of 0.82 and can return up to two similar examples. Similarity scores indicate the closeness of feature representations; they are not probabilities and do not confirm a biological diagnosis.

The expected FAISS index and corresponding label-map files were not found in the examined workspace. Therefore, image retrieval should be described as an optional source-code capability unless a populated index and successful test results can be demonstrated. The difference between bilinear resizing during index construction and nearest-neighbour resizing during inference should also be addressed when assessing reproducibility.

3.4 Measures and Evaluation Metrics

The evaluation framework uses classification metrics for disease detection and harvest readiness, regression metrics for bark-thickness prediction, and software tests to assess application integration. The selected metrics provide complementary information about predictive performance and system behaviors.

3.4.1 Accuracy

For multiclass classification, accuracy is calculated as the number of correct predictions divided by the total number of evaluated observations. In the binary formulation above, (TP), (TN), (FP), and (FN) represent true positives, true negatives, false positives, and false negatives, respectively.

3.4.2 Precision

Precision measures the proportion of positive predictions that are correct:

Higher precision indicates fewer false-positive predictions among the observations classified as positive.

3.4.3 Recall

Recall measures the proportion of reference positive observations correctly identified Higher recall indicates that fewer positive observations are missed by the model.

3.4.4 F1-Score

For the seven-class disease classifier, per-class and aggregate F1-scores should be reported where applicable. Macro F1 gives equal weight to each class, whereas weighted F1 accounts for class frequency. These measures help assess performance when the dataset contains unequal numbers of samples per class.

3.4.5 Confusion Matrix and ROC-AUC

A confusion matrix summarizes the relationship between reference labels and predicted labels across the seven disease classes or the harvest-readiness categories. It helps identify frequently confused conditions, including visually similar diseases, and errors involving unsupported inputs.

The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) measures a model's ability to distinguish a target class from other classes across different decision thresholds. For multiclass classification, the averaging strategy and implementation method should be specified. ROC-AUC should be reported alongside accuracy, precision, recall, and F1-score rather than used as a replacement for them. Its interpretation also depends on the evaluation dataset and the reliability of the reference labels.

3.4.6 Confidence Assessment

Evaluation of the 0.70 score and 0.15 margin thresholds should report the proportion of predictions flagged for review and the error rates among flagged and unflagged cases. These thresholds are operational heuristics, not proof of calibrated confidence.

3.4.7 Growth and Harvest-Readiness Evaluation

Growth prediction uses MAE, RMSE, and (R^2), reported in the target's units and linked to the evaluated model and test partition. The source of reference values must be stated.

Harvest readiness uses precision, recall, F1-score, and a confusion matrix. Decision thresholds should be reported because they affect false-positive and false-negative rates. Where labels are rule-derived, results demonstrate agreement with those rules rather than independently established field validity.

3.5 Data Analysis

Data analysis covers dataset quality, model training, predictive performance, uncertainty, and integrated application behaviour.

3.5.1 Dataset and Training Analysis

The analysis summarizes dataset sizes, class distributions, partitions, missing values, duplicates, exclusions, and target provenance. Disease-model training and validation records are examined alongside checkpoint-selection criteria. Equivalent records should be retained for growth and readiness models.

Differences between training and validation performance may indicate overfitting or insufficient learning, but these patterns must be interpreted alongside the data and experimental configuration. Reported procedures should be supported by actual training records.

3.5.2 Predictive Performance and Error Analysis

Disease-classification results are assessed using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC where available. Metrics must identify the exact model artifact and test subset. Results from an earlier checkpoint must not automatically be attributed to the exported model.

Growth predictions are compared with reference thickness values using MAE, RMSE, and (R^2). Readiness predictions are examined for false positives and false negatives. Misclassifications and large regression errors may be investigated in relation to image quality, symptom similarity, inconsistent labels, target ranges, or threshold choices; proposed explanations should be treated as hypotheses unless verified experimentally.

3.5.3 Uncertainty and Integrated-System Analysis

The analysis examines confidence indicators, flagged predictions, application errors, and behaviour near configured thresholds where records are available.

System integration is assessed at four levels:

- Correct execution of individual analytical models.
- Accurate transfer and presentation of model outputs.
- Correct application of decision and recommendation rules.
- Correct generation and interpretation of robotic guidance outputs, where implemented.

These levels distinguish software functionality from predictive reliability and physical harvesting performance.

3.6 Ethics, Privacy, Security, and Responsible AI

3.6.1 Data Integrity and Labelling

Class definitions, labeling procedures, and target provenance must be documented. Disease outputs should be described as image-based condition classifications rather than laboratory-confirmed diagnoses. Growth and readiness results must distinguish measured observations, expert judgments, and generated or rule-derived values.

3.6.2 Privacy and Security

Images may contain location metadata or incidental background information, so unnecessary information should not be retained. Where accounts and external data services are used, access should be restricted to authorized users and data should be transmitted and stored using appropriate security controls.

The backend should validate uploads, enforce file-size limits, protect endpoints, and handle errors without exposing credentials or sensitive internal information. Third-party vision services should be documented separately when enabled.

3.6.3 Responsible Interpretation and Generalization

CinnaAI is intended to support agricultural decision-making rather than replace expert judgment. Incorrect disease predictions may delay inspection, while inaccurate growth or readiness estimates may affect cultivation and harvesting decisions. Uncertain predictions should be clearly identified, and significant decisions should receive appropriate human review.

Performance may vary with plant variety, growth stage, environment, image quality, and symptom severity. Generalization claims therefore require representative and independent evaluation. Robotic guidance requires additional safeguards and physical testing; model predictions alone do not establish that a harvesting action is safe.

3.7 Reproducibility and Research Artefacts

3.7.1 Dataset Documentation

Dataset records should include sources, observation counts, class distributions, labeling methods, preprocessing, partitions, exclusions, and relevant acquisition conditions. Numerical datasets should also document variable definitions, units,

missing-value handling, and target generation. Real and synthetic data must be distinguished clearly.

3.7.2 Model and Training Artefacts

The main disease-classification artefacts include the EfficientNetB0 model, seven-class label mapping, training configuration, preprocessing procedure, backend implementation, and evaluation report.

The reviewed model file is `backend/cinnamon_multi_part_model.h5`. Its recorded SHA-256 hash is:

This hash identifies the reviewed file but does not establish dataset provenance or predictive accuracy. Growth and readiness models should be documented with their features, target definitions, configurations, and evaluation records.

3.7.3 Evaluation and Deployment Artefacts

Where available, reproducibility records should include dataset partitions, class counts, training logs, classification reports, confusion matrices, regression metrics, threshold settings, model identifiers, and integration-test results.

Software documentation should identify API endpoints, preprocessing, model-loading logic, frontend components, data services, and interfaces connecting prediction, decision support, and robotic guidance. Optional retrieval and external verification functions should be documented separately from the core classifier.

Reproducibility depends on access to the original data, reliable labels, model configurations, software dependencies, and evaluation records. The thesis should distinguish verified artefacts from experiments that have not been independently reproduced.

4 . 4. PRODUCT AND SOLUTION DESIGN

This chapter describes the design of the Cinnamon Leaf Disease and Condition Detection component of the CinnaAI Smart Cinnamon Monitoring and Harvest Decision-Support System. It covers system requirements, architecture, design alternatives, implementation, and professional practices. The component processes leaf images, classifies supported conditions, assesses prediction uncertainty, and presents results through the CinnaAI application.

The design combines multispectral image acquisition and processing with an EfficientNetB0-based classifier. The retained model accepts a three-channel RGB image resized to (224\times224) pixels. Therefore, multispectral processing and model input preparation are treated as separate stages.

4.1 Requirements

Requirements analysis defines the component's functional capabilities and quality expectations. Functional requirements describe system operations, while non-functional requirements address reliability, usability, performance, security, and maintainability.

4.1.1 Functional Requirements

- **FR1: Image Input and Validation** — Allow users to submit leaf images and validate file format, readability, size, and other applicable restrictions.
- **FR2: Multispectral Processing** — Support the implemented multispectral acquisition and processing workflow and prepare compatible image representations for classification.
- **FR3: Image Preprocessing** — Convert images to RGB, resize them to (224\times224), and construct the input tensor required by the retained model.
- **FR4: Image Classification** — Use the EfficientNetB0 model to classify images into seven categories: healthy cinnamon, leaf blight, leaf miner attack, fungal leaf patches, lower leaf gall, non-cinnamon, and upper leaf gall.
- **FR5: Prediction Results** — Identify the highest-scoring class and provide alternative predictions where required.
- **FR6: Uncertainty Assessment** — Flag predictions when the highest score is below 0.70 or the margin between the top two scores is below 0.15. These are operational thresholds, not validated diagnostic standards.
- **FR7: Result Presentation** — Return structured results distinguishing healthy, affected, unsupported, and review-required inputs, as applicable.
- **FR8: Application Integration** — Connect the frontend to the backend API and display prediction results, processing feedback, and errors.

- **FR9: Prediction Records** — Support storing and retrieving prediction records when the relevant services are configured, subject to access controls.
- **FR10: Error Handling** — Handle invalid uploads, image-decoding failures, inference errors, and service failures without presenting failed requests as successful predictions.

The classification labels represent image-level model outputs and should not be interpreted automatically as laboratory-confirmed diagnoses.

4.1.2 Non-Functional Requirements

- **NFR1: Reliability** — Return structured responses for valid requests and handle failures appropriately.
- **NFR2: Usability** — Present predictions, confidence indicators, and review messages clearly.
- **NFR3: Performance** — Provide practical response times for interactive use, assessed under the intended operating conditions.
- **NFR4: Maintainability** — Separate frontend interaction, API handling, preprocessing, inference, and result interpretation.
- **NFR5: Security** — Validate uploaded content and apply configured authentication and authorization controls to protected resources.
- **NFR6: Consistency** — Maintain the correct preprocessing operations, input dimensions, channel representation, class ordering, and thresholds.
- **NFR7: Reproducibility** — Document the model artifact, preprocessing configuration, label mapping, training settings, and evaluation procedure.
- **NFR8: Responsible Use** — Communicate prediction uncertainty and encourage expert review when necessary, without presenting outputs as definitive diagnoses.4.1.4 Requirement verification

The requirements should be verified through functional tests, integration tests, and model evaluation. Functional tests can confirm that valid images are accepted, invalid files are rejected, predictions are returned, and result categories are mapped correctly. Integration tests can verify communication between the frontend and backend and confirm the behaviour of configured storage and authentication services.

Model evaluation is different from functional verification. A system may successfully return a prediction while making an incorrect classification. Consequently, the operation of the software and the predictive performance of the model should be evaluated separately. Classification performance should be assessed using appropriate

metrics, including precision, recall, F1-score, and a confusion matrix, when the necessary labelled evaluation data are available.

4.2 Component Architecture within the Complete Solution

The disease-detection component operates within the CinnaAI application, which integrates plant monitoring and agricultural decision-support functions. The component classifies uploaded leaf images, while other modules handle growth prediction, harvest readiness, and related agricultural functions.

4.2.1 Architectural Overview

The application follows a frontend–backend architecture using Next.js, React, and TypeScript for the frontend and Python with FastAPI for the backend. The backend validates images, performs preprocessing, executes model inference, interprets predictions, and returns results.

The main architectural layers are:

1. **User interface:** Accepts images and displays predictions.
2. **API layer:** Validates requests and coordinates processing.
3. **Image-processing layer:** Prepares images for inference.
4. **Machine-learning layer:** Executes the EfficientNetB0 classifier.
5. **Result layer:** Interprets class probabilities and flags uncertain predictions.
6. **Supporting services:** Manage prediction records and other configured application services.

This separation supports maintainability and keeps model processing independent of the user interface.

4.2.2 Component Architecture Diagram

The architecture can be represented as:

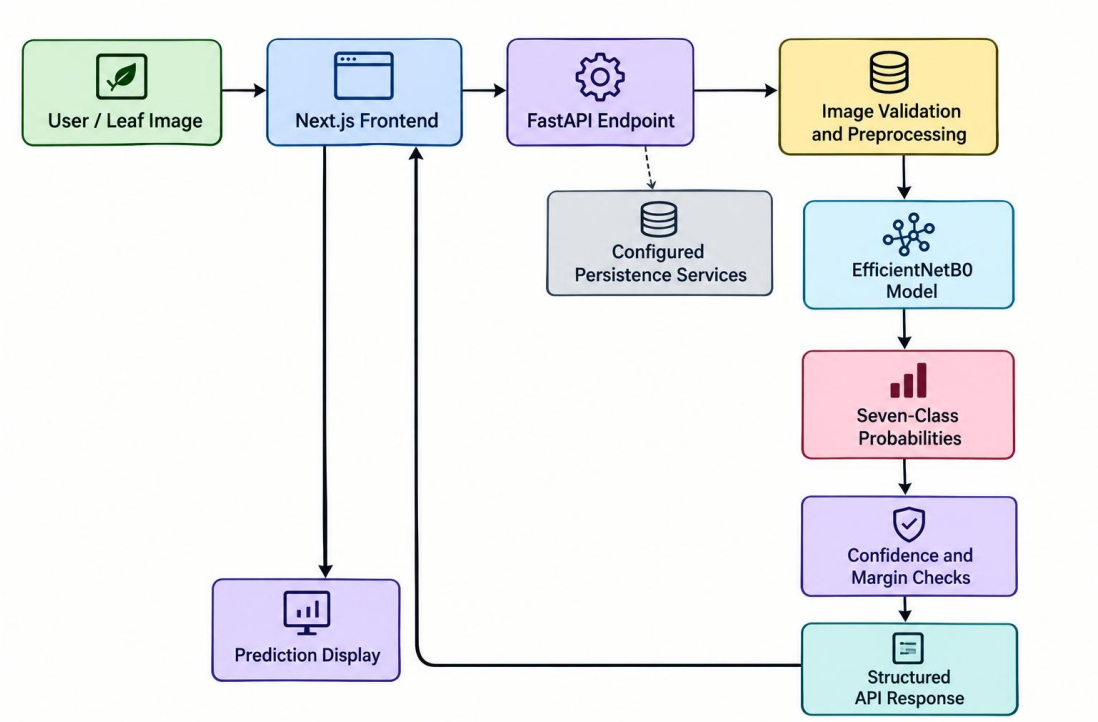


Figure 3 - Component Architecture Diagram

4.2.3 Frontend Layer

The frontend, built with Next.js, React, and TypeScript, provides image selection, submits prediction requests, manages processing states, and displays returned results. It communicates asynchronously with the backend and presents appropriate error messages when requests fail. Authentication-related components support protected routes, while classification remains the responsibility of the backend model.

4.2.4 Backend and API Layer

FastAPI coordinates the disease-classification workflow through the documented `/disease-predict` endpoint. It validates uploaded images, invokes preprocessing and inference, interprets model outputs, and returns structured responses. Centralized inference avoids duplicating model logic in the browser. Authentication and prediction-record management depend on the relevant services being configured.

4.2.5 Image Acquisition and Processing Layer

The documented inference pipeline converts uploaded images to RGB, resizes them to 224 × 224 pixels, and constructs a floating-point batch compatible with the retained model.

Although the project describes multispectral acquisition and processing, the EfficientNetB0 artifact accepts three-channel input. Therefore, the relationship between acquired spectral bands and the model's RGB input must be documented accurately. The classifier should not be described as directly processing multiple spectral bands unless this capability is implemented and verified.

Preprocessing must remain consistent between training, evaluation, and deployment, including channel format, resizing, and pixel scaling.

4.2.6 Machine-Learning Layer

The disease classifier uses the retained artifact `backend/cinnamon_multi_part_model.h5`, based on EfficientNetB0. Its architecture includes global average pooling, batch normalization, dropout, and a seven-unit softmax output layer.

The model receives a 224 × 224 × 3 image tensor and returns probabilities for seven classes. The backend identifies the highest-probability class and retains alternative predictions for uncertainty assessment. It performs image-level classification rather than lesion segmentation or direct disease-severity measurement.

4.2.7 Result Interpretation and Supporting Services

The backend applies two uncertainty checks: a maximum class probability below 0.70 or a difference between the two leading probabilities below 0.15 flags a prediction as weak. These thresholds indicate operational uncertainty but do not guarantee correctness.

Prediction history may be stored through Firestore or other configured services. Optional retrieval or external-verification functions should only be described as active diagnostic features when their operation has been verified.

4.2.8 Integration with CinnaAI

The disease classifier shares the wider application's interface and backend but retains a distinct responsibility: classifying uploaded leaf images. It does not directly predict bark thickness, determine harvest readiness, or control robotic harvesting. Any future

integration of its output into broader agricultural decisions requires explicit rules and appropriate validation.

4.3 Design Alternatives and Decisions

4.3.1 Automated Classification versus Manual Inspection

Manual inspection allows agricultural specialists to interpret symptoms in context but can require time and vary between observers. Automated classification provides repeatable initial assessments.

4.3.2 Conventional Image Processing versus Deep Learning

Traditional methods use handcrafted colour and texture features, whereas CNNs learn visual representations from labelled images. The latter can capture complex patterns but depend on suitable training data.

Decision: Retain EfficientNetB0 for the seven-class classification task.

4.3.3 EfficientNetB0 versus Alternative Architectures

Architectures such as MobileNet may offer advantages in model size and inference speed. However, changing architectures requires consistent training and evaluation.

Decision: Use the retained EfficientNetB0 artifact and consider alternatives in future comparative experiments.

4.3.4 Multispectral Input versus Three-Channel Input

A model accepting multiple spectral bands requires a compatible architecture and training pipeline. Alternatively, spectral data may be transformed into a three-channel representation, provided the transformation is suitable and documented.

Decision: Retain the existing three-channel classifier interface. Document and evaluate the actual transformation from acquired data to model input before claiming multispectral benefits.

4.3.5 Backend versus Browser Inference

Browser inference may reduce image uploads but requires client-compatible models and sufficient device resources. Backend inference centralizes preprocessing, model loading, and prediction logic.

4.3.6 Fixed Predictions versus Uncertainty-Aware Outputs

Always presenting the highest-probability class may encourage overconfidence. Confidence and probability-margin checks can flag potentially unreliable predictions.

Decision: Retain uncertainty checks, but validate the thresholds against labelled data before treating them as reliable decision boundaries.

4.4 Detailed Design

4.4.1 Image Input and Validation

The workflow begins when a user uploads a leaf image. The backend checks that the upload contains valid, decodable image data and complies with applicable file and dimension restrictions. Invalid uploads should produce appropriate error responses rather than classification results.

4.4.2 Image Preprocessing

The documented inference workflow consists of:

1. Receiving and validating the uploaded image.
2. Decoding the image.
3. Converting it to RGB.
4. Resizing it to 224 × 224 pixels using nearest-neighbour interpolation.
5. Creating a floating-point array and batch dimension.
6. Passing the prepared image to the model.

The documented input uses pixel values in the 0–255 range. Additional normalization should not be introduced unless the model's training pipeline requires it.

The precise handling of multispectral bands must be documented from the actual implementation. Channel selection, transformation, resizing, and pixel scaling must remain compatible with the retained model.

4.5 Standards and Professional Practice

The component should follow sound software engineering, machine-learning, security, and responsible-AI practices. This discussion describes relevant practices rather than claiming formal certification.

4.5.1 Software Engineering

The frontend, API, preprocessing, inference, and result interpretation are separated into distinct responsibilities. Clear interfaces, version control, documented configuration, and integration testing support maintainability. Model versions should be recorded so predictions can be associated with the relevant artifact and preprocessing pipeline.

4.5.2 Data Quality and Machine Learning

Training data should contain reliable labels and representative images. Duplicate images across training and test sets should be avoided to reduce data leakage. Training, validation, and testing must serve separate purposes, and preprocessing settings, class mappings, and augmentation methods should be documented. Class imbalance should also be assessed.

4.5.3 Model Evaluation and Reporting

Evaluation should include a confusion matrix and suitable class-level metrics, such as precision, recall, and F1-score. Dataset sources, sample counts, class distributions, and splitting procedures should be reported where available. Software functionality must be distinguished from predictive accuracy, and unverified metrics should not be presented as established results.

Confidence thresholds should be validated against labelled data. The model's limitations, including its inability to segment lesions or directly measure disease severity, should also be acknowledged.

4.5.4 Security and Privacy

Uploaded files should be validated, protected endpoints should enforce backend authentication and authorization, and credentials should be stored securely. Access to images and prediction records should follow appropriate permissions. Data-retention and deletion procedures should be documented where relevant.

4.5.5 Responsible AI and Human Oversight

The classifier supports agricultural decisions but does not replace expert judgement. Uncertain predictions should be flagged for review, and performance should be assessed across relevant image conditions where data are available. Output labels such as `leaf_miner_attack`, `lower_leaf_gall`, and `upper_leaf_gall` should be presented as model classifications rather than confirmed biological diagnoses.

4.5.6 Reproducibility and Documentation

The project should preserve the model artifact, class-label mapping, preprocessing code, training configuration, and evaluation procedure. Implemented features must be distinguished from optional or incomplete integrations. Multispectral acquisition details, sensor specifications, spectral bands, and transformation methods should be documented when available rather than inferred or fabricated.

5. IMPLEMENTATION, INTEGRATION AND TESTING

This chapter describes the implementation, integration, testing, and deployment of the Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance component within the CinnaAI Smart Cinnamon Monitoring and Harvest Decision-Support System. The component combines machine learning, image processing, environmental data handling, and web-based software technologies to support three main tasks: cinnamon growth estimation, disease detection, and harvest-readiness assessment.

The implementation uses a web-based frontend to collect user inputs and display results, while a Python-based backend manages prediction requests and connects them to the relevant machine learning models and supporting services. The growth prediction function estimates cinnamon bark thickness, the disease detection function classifies uploaded leaf images, and the harvest-readiness function generates a decision-support output based on selected plant indicators. The resulting information is intended to assist users in making more informed monitoring and harvesting decisions.

This chapter also discusses integration between the three functions and the wider CinnaAI platform. Particular attention is given to the distinction between successful software execution, model performance, and the physical operation of a robotic harvesting mechanism. These represent different levels of implementation and require different forms of verification.

5.1 Development Environment and Technology Stack

The CinnaAI development environment combines Python-based machine learning and backend services with a modern web frontend. This architecture supports image processing, prediction execution, data exchange, authentication, and hardware-interface integration.

5.1.1 Programming Languages and Development Environment

Python supports backend development, data processing, and machine learning. TypeScript and React are used to develop the frontend, while Next.js provides the application framework and routing. Separating the frontend and backend allows independent development while maintaining communication through defined API endpoints.

5.1.2 Technology Stack

Table 3 - summarizes the principal technologies

Technology	Role
Python	Backend and machine learning
FastAPI and Uvicorn	API development and server execution
TensorFlow/Keras	EfficientNetB0 inference
Scikit-learn	Regression, classification, and evaluation
NumPy and pandas	Numerical and tabular data processing
Pillow	Image decoding and preprocessing
Next.js, React, TypeScript	Frontend development
Firebase Authentication	User authentication and token validation
Firebase Firestore	Prediction-record storage where configured
Firebase Realtime Database	Environmental telemetry where configured
ESP32 and serial communication	Supported hardware-control operations
Environment configuration	Application settings and credentials

TensorFlow/Keras supports disease classification, while Scikit-learn supports growth prediction and harvest-readiness classification. FastAPI provides a common backend interface, and Next.js delivers the user interface. OpenCV or alternative technologies should only be listed as implemented components if their use is verified.

5.1.3 Machine Learning and Data Processing

CinnaAI uses separate models for three tasks: Random Forest regression for bark-thickness estimation, EfficientNetB0 for leaf disease classification, and a classification model for harvest-readiness assessment. The backend manages preprocessing, model execution, and result interpretation, allowing models to be updated independently of the frontend.

5.1.4 Supporting Services and Configuration

Firebase Authentication supports sign-in and identity-token validation. Firestore can store prediction records, while Realtime Database can supply environmental readings when configured. A serial communication interface supports selected commands to an ESP32-based control arrangement, but this does not establish successful autonomous harvesting. Credentials and service settings should be managed securely outside publicly exposed source code.

5.2 Component Implementation

The component comprises three machine learning functions: cinnamon growth prediction, leaf disease detection, and harvest-readiness assessment. Each accepts user or sensor data, validates and processes the input, executes the appropriate model, and returns results for display.

5.2.1 Cinnamon Growth Prediction

The growth prediction function estimates bark thickness using plant age, temperature, humidity, and soil moisture. Environmental readings may also be retrieved from a configured telemetry service. The backend arranges the input features and passes them to a Random Forest regressor, which produces an estimated bark thickness in millimeters.

$$G = \mathrm{clip}\left(100\ \frac{\hat{b} - 0.5}{1.0}, 0, 100\right)$$

where:

- $\hat{b}$ is the predicted bark thickness (mm).
- G is the growth indicator (0–100).
- $\mathrm{clip}(x, a, b)$ limits the value of x to the range $[a, b]$.

The indicator is a transformed application value, not a direct measure of biological maturity or harvesting success. Results are displayed with a corresponding status. Their reliability depends on training-data quality, target construction, and independent validation against measured bark thickness.

5.2.2 Cinnamon Leaf Disease Detection

The disease detection function uses an EfficientNetB0-based CNN to classify uploaded images. The backend validates the uploaded file, converts it to RGB, resizes it to (224 \times 224) pixels, and prepares the input tensor according to the model's requirements.

Although multispectral acquisition and processing form part of the broader image workflow, the retained classifier accepts a three-channel input. Therefore, multispectral processing must be distinguished from the final RGB representation supplied to the model.

The classifier predicts one of seven classes:

1. Healthy cinnamon.
2. Leaf blight.
3. Leaf miner attack.
4. Leaf patches fungal.
5. Lower leaf gall.
6. Non-cinnamon.
7. Upper leaf gall.

The backend returns the leading class and relevant confidence information. Predictions are flagged for review when the highest class probability is below 0.70 or the difference between the two highest probabilities is below 0.15. These thresholds identify potentially uncertain results but do not guarantee correctness.

The frontend displays the predicted category and review information. The model performs image-level classification; it does not directly segment lesions, measure affected leaf area, or confirm the biological cause of symptoms.

5.2.3 Harvest-Readiness Assessment

The harvest-readiness function accepts selected plant features and a plant identifier. After validating the input, the backend passes the features to the configured classifier and returns a readiness-related category or action label.

Unlike the bark-thickness regressor, this model produces a classification output based on its own input features. Its reliability depends on the representativeness of its training data and labels. A prediction alone cannot confirm biological readiness, successful bark peeling, or safe robotic harvesting. Independent validation against actual harvesting outcomes is required before operational use.

5.2.4 Robotic Harvest Guidance Interface

CinnaAI includes image-measurement support, cut-depth-related guidance, and an ESP32 serial interface for supported relay commands. These functions provide software and hardware-interface capabilities but do not demonstrate a fully autonomous harvesting system.

Operational harvesting would require validated plant localization, accurate measurements, suitable cutting parameters, safe actuator control, collision avoidance, emergency stopping, and physical testing. Displayed recommendations, dispatched commands, and completed physical operations must therefore be verified separately.

5.2.5 Prediction Output Handling

The backend returns structured results for frontend display. Growth outputs include estimated bark thickness and the growth indicator; disease outputs include the predicted class and confidence information; harvest-readiness outputs include the relevant category or decision label.

Invalid images, missing inputs, and inference failures should produce appropriate error responses. Prediction records and email notifications may be processed through configured services, but their success must be verified independently of model inference.

5.3 Integration with Other Components

The three machine learning functions share a frontend and backend architecture, enabling consistent input collection, prediction requests, and result presentation.

5.3.1 Frontend and Backend Integration

The Next.js frontend communicates with the FastAPI backend through HTTP requests and designated endpoints. The frontend collects inputs and displays request status and results, while the backend performs authoritative validation, model execution, and interpretation. This separation reduces frontend dependence on model-specific implementation details.

5.3.2 Growth Prediction and Environmental Monitoring

The growth function uses manually entered environmental values or readings retrieved from Firebase Realtime Database when configured. Retrieved values must match the model's expected features. Missing, outdated, or invalid readings should be handled explicitly because prediction quality depends on both model performance and input reliability.

5.3.3 Disease Detection and Supporting Services

The disease detection function uses the shared authentication and backend infrastructure. Users submit images and receive classification results through the web interface. Firestore storage and notifications may be available when configured. Optional retrieval or external-model functions should only be described as active parts of inference when their execution has been verified.

5.3.4 Growth and Harvest-Readiness Integration

The growth model estimates bark thickness and derives a threshold-based indicator, while the harvest-readiness classifier produces a separate classification. These results can support joint review but are not interchangeable. A growth indicator labelled "Ready to Harvest" does not confirm the separate classifier's output or establish that harvesting is appropriate.

5.3.5 Robotic Guidance and Hardware Integration

Plant measurements and readiness results may inform potential harvesting decisions, while the ESP32 interface provides a separate route for supported control commands. Physical actions should not be triggered solely by machine learning predictions; suitable safety checks, hardware-state validation, and human supervision are required.

The available implementation does not establish complete operational integration for all robotic-service endpoints. CinnaAI should therefore be described as providing software guidance and selected control-interface functions rather than a fully validated autonomous harvesting solution.

5.3.6 Data Storage and Authentication

Firebase Authentication supports user sign-in and identity-token validation for protected operations. Prediction records may be stored in Firestore or local CSV files, depending on the configured workflow. Storage should be tested independently because successful inference does not guarantee successful record persistence. The application should distinguish prediction success from storage failure.

5.4 Verification and Software Testing

Testing evaluates functional correctness, input handling, integration, model performance, and hardware-interface behaviour. These areas must be assessed separately: successful API execution does not prove prediction accuracy, and model performance does not establish safe physical harvesting.

5.4.1 Testing Strategy

The verification strategy covers:

1. **Functional testing:** verifies that each feature performs its intended operation.

2. **Input-validation testing:** checks handling of missing, invalid, or unsupported inputs.
3. **Model-interface testing:** verifies model input formats and output structures.
4. **Integration testing:** evaluates communication among the frontend, backend, models, and supporting services.
5. **Model evaluation:** measures predictive performance using appropriate labelled evaluation data.
6. **Hardware-interface testing:** checks serial communication and control functions separately from physical harvesting.
7. **Deployment testing:** verifies configuration, startup, connectivity, and service availability.

This strategy assesses individual functions and overall system integration while distinguishing software correctness from real-world agricultural and robotic performance.

5.4.2 Functional and input-validation tests

Functional testing checks that valid inputs can be submitted and that the application returns the expected type of result. Input-validation testing checks that the system handles missing or malformed inputs without causing an uncontrolled failure.

Table 5.3 presents the main test cases for the component. These cases form a verification plan; individual outcomes should be marked as passed or failed only after the corresponding tests have been executed and recorded.

Table 4 - Test Cases

Test ID	Function	Test case	Expected result
T01	Growth prediction	Submit valid plant and environmental inputs	The backend returns an estimated bark-thickness value
T02	Growth prediction	Submit missing or invalid feature values	The request is rejected or handled with a clear validation response
T03	Disease detection	Upload a valid supported leaf image	The backend returns a classification response
T04	Disease detection	Upload an empty or unreadable file	The backend rejects the input without presenting a prediction
T05	Disease detection	Submit an image that violates configured limits	The backend returns an appropriate validation error

T06	Disease detection	Check the model output mapping	The seven output positions map to the correct class labels
T07	Disease detection	Submit a prediction with weak confidence or margin	The result contains the appropriate review indication
T08	Harvest readiness	Submit valid required plant features	The backend returns a readiness-related prediction
T09	Harvest readiness	Submit missing or invalid required features	The backend returns an appropriate validation response
T10	Authentication	Access a protected endpoint without valid credentials	Access is denied according to the configured authentication rules
T11	Data storage	Generate a prediction while storage is available	The configured persistence operation is attempted and its result is handled
T12	Data storage	Simulate an unavailable storage service	The application handles the storage failure without falsely reporting successful persistence
T13	Frontend	Submit a valid prediction request	The interface displays the returned result
T14	Frontend	Simulate a backend failure	The interface displays an error or failure state
T15	Hardware interface	Send a supported command through the serial interface	The communication outcome is checked separately from physical harvesting

Table 5.3: Functional and integration test cases

The tests should be executed using valid, invalid, and boundary-case inputs. For example, the disease detection workflow should be tested with supported images and files that cannot be decoded. The growth workflow should be tested with valid numerical values and inputs outside the accepted range. The harvest-readiness workflow should be tested with complete and incomplete feature sets.

The expected result should be defined before testing. This makes it possible to compare actual system behaviour with the requirement rather than deciding after execution whether a result is acceptable.

5.4.3 Model-Interface Verification

Model-interface verification checks whether each prediction function supplies valid inputs and correctly interprets model outputs. For growth prediction, tests verify feature ordering, output units, growth-indicator calculations, and thresholds.

For disease detection, verification checks the (224 \times 224 \times 3) RGB input, floating-point representation, seven-class probability vector, and class-label mapping. Confidence and top-two probability-margin checks should also be tested against their configured thresholds.

For harvest readiness, tests verify feature ordering, output interpretation, and mapping to the appropriate decision-support label. These checks confirm interface correctness but do not establish real-world model generalization.

5.4.4 Evaluation of Growth Prediction

Growth prediction should be evaluated using (R^2), mean absolute error (MAE), and root mean squared error (RMSE).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Here, (y_i) represents the reference bark thickness, (\hat{y}_i) the predicted value, and (\bar{y}) the mean reference value. MAE measures average absolute error, while RMSE penalizes larger errors more heavily.

Evaluation should document the data source, target definition, sample count, and data-splitting procedure. If targets are generated using predefined formulas rather than independent physical measurements, the results demonstrate agreement with those rules, not verified field accuracy.

5.4.5 Evaluation of Disease Detection

Disease detection should be evaluated using labelled images excluded from training. A confusion matrix, class-wise precision, recall, and F1-score can identify

classification strengths and weaknesses. Overall accuracy may also be reported, but it should not be the sole metric because class imbalance can conceal poor performance on less frequent classes. ROC-AUC may be included where an appropriate multiclass evaluation method is used.

Evaluation should identify the tested model artifact and data split and describe how closely the images represent actual operating conditions. If multispectral processing is a central contribution, the processed inputs and suitable baseline comparisons should also be documented.

5.4.6 Evaluation of Harvest-Readiness Prediction

The harvest-readiness classifier should be evaluated using precision, recall, F1-score, and a confusion matrix, together with the evaluation-data source, class distribution, and applicable decision thresholds.

The origin of the reference labels is important. If labels are generated from predefined rules, the metrics indicate how well the model reproduces those rules rather than how accurately it predicts actual harvesting outcomes. Practical validation requires independent plant-readiness assessments and, where applicable, observed bark-peeling or harvesting results.

Readiness predictions should remain decision-support information, not direct authorization for robotic cutting. Physical harvesting requires additional operational and safety checks.

5.4.7 Integration Testing

Integration testing verifies communication among the frontend, backend, models, authentication services, databases, environmental telemetry, and hardware interfaces. Tests should confirm that inputs are transferred and interpreted correctly and that outputs are returned or stored as expected.

For example, environmental readings should map to the correct growth-model features, uploaded images should be preprocessed and classified successfully, and saved records should be retrievable where persistence is configured. Optional retrieval services, external-model verification, notifications, and database operations must be tested independently rather than assumed to work whenever inference succeeds.

5.4.8 Hardware and Robotic-Interface Verification

Hardware-interface testing should verify serial connection establishment, command formatting, dispatch, and communication-error handling. Successful command transmission does not prove correct actuator movement or successful harvesting.

Physical testing should assess positioning and cut-depth accuracy, repeatability, collision avoidance, emergency stopping, and safe behaviour when sensor data or predictions are unavailable. Controlled trials are necessary before real harvesting. If physical trials have not been completed, the report should describe the implemented software and interfaces without claiming validated autonomous harvesting.

5.4.9 Test Reporting and Limitations

Test records should include the test identifier, input conditions, expected and actual results, pass/fail status, and observations. Executed tests must be distinguished from proposed tests, and model scores should be linked to the specific artifact, dataset, and evaluation procedure.

Discrepancies should be documented and corrected. Software functionality, predictive performance, and suitability for real agricultural and robotic conditions must be evaluated separately.

5.5 Deployment and Operation

CinnaAI deployment involves configuring the frontend, backend, machine learning models, and supporting services. Successful operation requires the correct model artifacts, dependencies, environment settings, access controls, and service connections.

5.5.1 Backend Deployment

The FastAPI backend runs through Uvicorn and requires the Python dependencies used for data processing, machine learning, image handling, authentication, and configured services. Environment variables and credentials must be configured before startup, and model artifacts must be accessible at the expected paths.

The disease model is stored at `backend/cinnamon_multi_part_model.h5`. Deployment should verify that it loads successfully and that its input dimensions and

class mapping match the inference code. Growth and harvest-readiness models must also be available in the formats expected by their respective functions.

If models are trained during startup, the training-data source and startup procedure should be documented because they affect execution time and reproducibility. Appropriate logging, error handling, and production settings should be used, without exposing debugging configurations or development credentials.

5.5.2 Frontend Deployment

The Next.js, React, and TypeScript frontend requires its dependencies, backend API address, and other application settings to be configured before deployment. The frontend must communicate with the backend successfully in the target environment.

Cross-origin settings, authentication, and failed-request handling should be tested. The interface should provide useful feedback when services are unavailable, and browser-exposed configuration must not contain private credentials.

5.5.3 Deployment of Supporting Services

Firebase Authentication, Firestore, and Realtime Database require valid project settings and appropriate access rules. Authentication tokens must be validated, and data operations must use suitable permissions.

Environmental monitoring depends on accurate, available telemetry. Missing or outdated readings should be clearly identified rather than presented as current measurements. Email notifications, retrieval functions, and external-model verification should be documented according to their actual configuration and availability.

The ESP32 interface requires an accessible serial connection and compatible hardware settings. The controller and connected actuators must be checked before commands are issued, with hardware access restricted to authorized operations and appropriate safety measures in place.

6 RESULTS AND VALIDATION

This chapter presents the research results, technical performance, and integration outcomes of the Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance component within the CinnaAI system. The evaluation focuses on three principal machine-learning functions: cinnamon bark-thickness prediction, cinnamon leaf disease and condition classification, and harvest-readiness classification. It also discusses the integration of these functions into wider application and the evidence available for validating their practical usefulness.

The results are considered at two levels. Research results describe the predictive performance of the machine-learning models using the evaluation metrics available. Product and technical results describe the implementation of the models and their connection to the application. These two levels are related but not interchangeable: a model may achieve satisfactory evaluation metrics while still requiring further testing under real-world operating conditions.

The chapter uses the performance values shown in the supplied model-evaluation figures. Formal stakeholder-validation results and evidence of completed physical robotic harvesting trials were not available in the supplied evaluation material. Consequently, the chapter distinguishes measured model results from implementation outcomes and validation activities that require additional evidence.

6.1 Research results

The research component investigates the use of machine learning and image-processing techniques to support cinnamon growth monitoring, leaf disease detection, and harvest-readiness assessment. Each task has a different prediction objective and therefore requires suitable evaluation metrics.

The growth prediction model produces a continuous numerical estimate of bark thickness. Disease detection produces a class prediction from an input image. Harvest-readiness assessment produces a classification result based on the input features and the decision policy applied by the model. The results for these tasks are discussed separately below.

6.1.1 Cinnamon growth prediction results

The growth prediction function uses a Random Forest regression model to estimate cinnamon bark thickness. The supplied evaluation figure presents an actual-versus-predicted-style scatter plot, a residual plot, and a comparison with reference estimators. The reported Random Forest performance is summarized in Table 6.1.

Table 5 - growth prediction evaluation results

Evaluation metric	Reported result
Coefficient of determination ((R^2))	0.8393
Root mean squared error (RMSE)	0.1648 mm
Mean absolute error (MAE)	0.1317 mm
Mean squared error (MSE)	0.0272 mm²
Number of paired evaluation records shown	2,000

The reported (R^2) value of 0.8393 indicates that the model accounts for a substantial proportion of the variation in the evaluation target relative to a mean-based reference. The RMSE of 0.1648 mm represents the square root of the average squared prediction error and is expressed in the same unit as bark thickness. The reported MAE of 0.1317 mm represents the average absolute difference between the model predictions and reference values.



Figure 4 - Scatter Plot of Growth Prediction

The scatter plot shows a general relationship between the reference values and predicted bark thickness. However, the distribution of points also indicates that individual predictions differ from the reference values. The residual plot provides additional information about the size and distribution of these differences. Residual analysis is important because a model can achieve a reasonably high (R^2) while still exhibiting systematic errors over particular ranges of the target variable.

The supplied figure also compares the Random Forest model with reference estimators. The displayed results report an (R^2) of 0.8677 for a direct formula-based estimator and a negative (R^2) of -0.0006 for a training-target-mean estimator. The formula-based estimator therefore has a higher reported (R^2) than the Random Forest in this comparison. This is an important finding: the results do not establish that the Random Forest is the best-performing estimator among all methods shown. The selection of the final approach should consider the error metrics, the meaning of each reference estimator, and the intended use of the prediction.

The growth prediction results suggest that the Random Forest model can estimate the evaluation target with measurable accuracy. Nevertheless, the results should not be interpreted as proof of equivalent accuracy across different plantations, cinnamon varieties, environmental conditions, or measurement procedures. Independent evaluation using representative field observations is needed to establish the model's generalization performance.

6.1.2 Cinnamon leaf disease and condition detection results

The disease detection function uses an EfficientNetB0-based convolutional neural network to classify cinnamon leaf images into seven output categories. These categories are healthy cinnamon, leaf blight, leaf miner attack, leaf patches fungal, lower leaf gall, non-cinnamon, and upper leaf gall.

The supplied evaluation figure compares the initial EfficientNetB0 checkpoint with the exported final model. The reported metrics are presented in Table 6.2.

Table 6 - disease-classification performance

Model checkpoint	Precision	Recall	F1-score	ROC-AUC
EfficientNetB0 initial checkpoint	84.07%	86.11%	84.65%	0.9860
EfficientNetB0 exported final model	78.43%	80.82%	78.60%	0.9824

The initial checkpoint achieves a precision of 84.07%, a recall of 86.11%, and an F1-score of 84.65%. Its ROC-AUC is reported as 0.9860. The exported final model achieves a precision of 78.43%, a recall of 80.82%, and an F1-score of 78.60%, with a ROC-AUC of 0.9824.

The results show a reduction in precision, recall, and F1-score between the two evaluated checkpoints. The F1-score decreases by 6.05 percentage points, from 84.65% to 78.60%. ROC-AUC decreases by 0.0036, from 0.9860 to 0.9824.

The relatively high ROC-AUC values indicate strong class-discrimination performance under the evaluation procedure used to produce the supplied figure. However, ROC-AUC does not by itself show whether every disease class is recognized reliably. It is therefore important to consider precision, recall, class-specific performance, and the confusion matrix alongside ROC-AUC.

The difference between the initial and exported final checkpoints is particularly important from a product perspective. The initial checkpoint has better reported classification metrics, while the exported final model represents the artifact intended for application inference. The final deployed artifact should be evaluated directly, because performance from an earlier checkpoint cannot automatically be attributed to the model used by the application.

The supplied figure describes the results as seven-class classification, with precision, recall, and F1-score summarized using macro averaging and ROC-AUC calculated using a micro one-versus-rest approach. These aggregation methods should be retained when reporting the metrics, since results calculated using different averaging approaches may not be directly comparable.

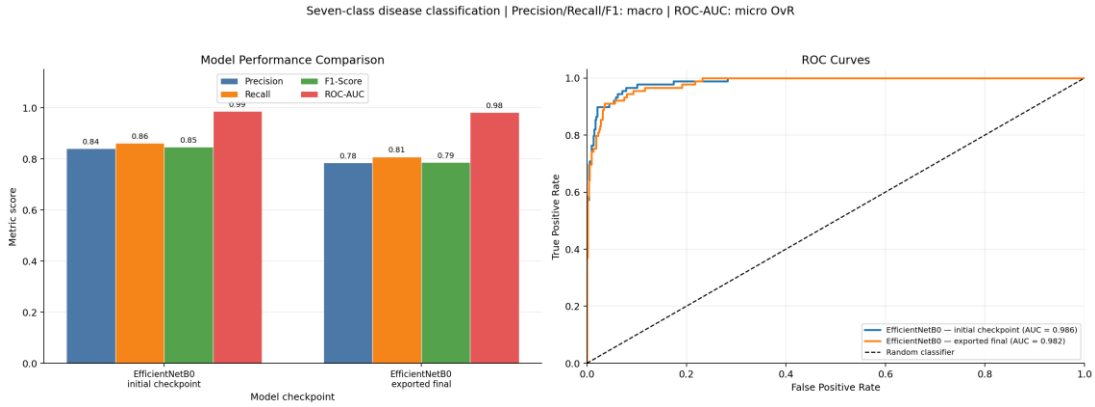


Figure 5 - Disease classification

The results support the feasibility of using a convolutional neural network for cinnamon leaf image classification. They do not establish that every prediction represents a confirmed disease diagnosis. In practical use, classification errors may result from visual similarity between conditions, image quality, environmental variation, or differences between the evaluation images and images captured in the field.

The component's confidence and prediction-margin checks provide an additional mechanism for flagging uncertain outputs. However, the supplied performance figures do not independently establish that the confidence thresholds are optimally calibrated.

Further evaluation should examine class-specific errors and the frequency with which uncertain predictions are correctly flagged.

6.1.3 Harvest-readiness prediction results

The harvest-readiness function uses a classifier to estimate whether a plant meets a defined readiness condition. The supplied evaluation figure compares two decision policies applied to the same classifier: a binary decision threshold of 0.50 and a stricter harvest-approval threshold of 0.80.

Table 7 - Harvest-classification results

Metric	Binary decision threshold (0.50)	Harvest-approval threshold (0.80)
Precision	100.00%	100.00%
Recall	92.31%	46.15%
F1-score	96.00%	63.16%
ROC-AUC	1.000	1.000
Average precision, as shown in the figure	1.000	1.000

At the binary decision threshold of 0.50, the classifier reports precision of 100.00%, recall of 92.31%, and an F1-score of 96.00%. Under the stricter harvest-approval threshold of 0.80, precision remains at 100.00%, but recall decreases to 46.15%, and the F1-score decreases to 63.16%.

The difference demonstrates the effect of changing the decision threshold. At the stricter threshold, fewer observations qualify for harvest approval. This may reduce the number of positive approvals, but it also means that more reference-positive cases are not approved. The lower recall therefore represents a trade-off between restrictive approval and the ability to identify all cases labelled as ready.

The precision values of 100% mean that no false-positive predictions were recorded among the positive predictions evaluated under the respective policies. This finding should be interpreted in the context of the evaluation sample and its reference labels. It does not establish that false approvals will never occur in field conditions.

The ROC-AUC and average-precision values of 1.000 are also reported in the supplied figure. Such values indicate perfect separation under the evaluation procedure represented by the figure. They require particular care in interpretation because the evaluation data, reference-label construction, class distribution, and possible overlap with the training data influence the reliability of these estimates. Independent testing

on representative, previously unseen examples is necessary before the result can be generalized to operational harvesting decisions.

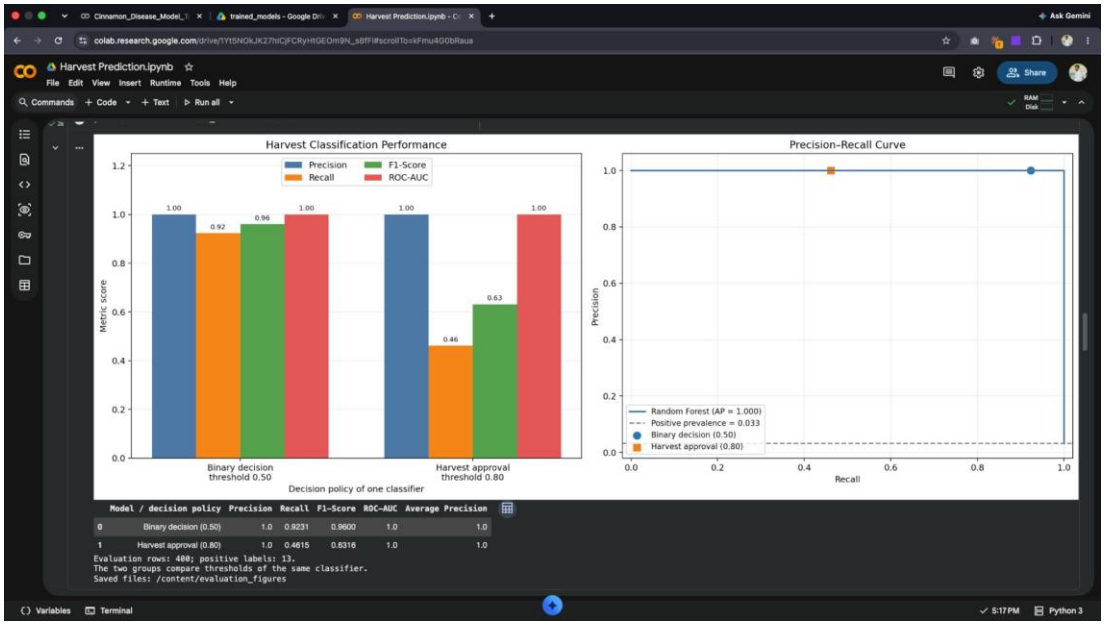


Figure 6 - Harvest Classification

The harvest-readiness results show that decision thresholds substantially affect the balance between precision and recall. A suitable threshold should be selected according to the practical consequences of false approval and missed readiness. For robotic harvesting, this decision should also include independent safety checks rather than relying exclusively on the classifier output.

6.2 Product and technical results

The product and technical results concern the implementation of the three machine-learning functions and their availability within the CinnaAI application. They address whether the system provides the intended software workflow and whether each model can be used through application architecture.

6.2.1 Implementation of the growth prediction function

The growth prediction function connects plant and environmental input values to a Random Forest regression model. The model produces a bark-thickness estimate, which the application can transform into a growth indicator and a descriptive status.

The technical result is a workflow that connects input collection, model inference, and result presentation. The numerical prediction gives the user a quantitative estimate rather than only a qualitative assessment of plant growth.

The meaning of the displayed growth indicator must remain clear. It is derived from the model's bark-thickness estimate using application logic and is not a direct measurement of plant maturity. The usefulness of the indicator therefore depends on the model's evaluation performance and the suitability of the thresholds used to interpret the estimate.

6.2.2 Implementation of disease detection

The disease detection function connects image upload and preprocessing with the retained EfficientNetB0 classifier. The model accepts an input representation of 224 × 224 pixels with three channels and produces a probability output over seven classes.

The backend maps the model output to the relevant class label and applies the configured confidence and top-two margin checks. These checks support the identification of predictions that should be reviewed rather than accepted without further inspection.

The product result is an image-based classification workflow integrated with the application. The output supports preliminary identification of a leaf condition but should not be treated as laboratory confirmation of a disease or as a substitute for expert assessment where such confirmation is required.

Multispectral image acquisition and processing are part of the implemented workflow described for this research. The final retained classifier nevertheless has a three-channel input interface. Accordingly, the thesis should explain the actual processing steps that convert the acquired and processed image information into the representation supplied to the classifier. This is necessary to make the multispectral contribution reproducible and to distinguish the acquisition stage from the model's final input format.

6.2.3 Implementation of harvest-readiness assessment

The harvest-readiness function uses plant-related input features to produce a classification result. The result can be displayed as decision-support information and considered alongside growth estimates and other observations.

The threshold comparison shows that the application's harvest decision policy can be made more restrictive by raising the approval threshold. The reported evaluation demonstrates the associated reduction in recall and F1-score under the stricter policy.

This technical capability allows the decision policy to reflect the relative importance of approving only likely-ready plants versus identifying as many ready plants as possible. The threshold should be chosen using representative validation data and the consequences of incorrect decisions. It should not be selected solely because one threshold produces a higher score on a particular evaluation set.

6.2.4 Web application and supporting services

The three prediction functions are supported by a common software architecture that uses a Next.js frontend and a FastAPI backend. The frontend collects inputs and presents the returned results, while the backend validates requests and invokes the appropriate model.

The architecture separates user interaction from machine-learning inference. This separation supports maintainability because model logic can be updated without requiring all prediction operations to be implemented in the frontend.

The system also contains supporting service integrations for authentication and configured data operations. Their operational status must be assessed separately from model execution. For example, successful inference does not necessarily mean that the result was stored, a notification was sent, or an environmental reading was retrieved successfully.

The product outcome is therefore an integrated application structure that brings the machine-learning functions together through a shared interface. Full operational reliability still depends on verification of individual endpoints, service connections, configuration, and error-handling behaviour.

6.3 User or Stakeholder Validation

Stakeholder validation assesses whether CinnaAI's outputs are understandable, relevant, and useful to intended users, including cinnamon growers, agricultural specialists, field officers, and technical personnel. Model performance and user acceptance are separate forms of evidence: accurate predictions may be difficult to interpret, while a usable interface does not guarantee model reliability.

6.3.1 Validation Objectives

Validation should determine whether users can:

- Enter plant and environmental data for growth prediction.
- Upload images and interpret disease-classification results.
- Recognize uncertain predictions that require further review.
- Understand harvest-readiness outputs and decision thresholds.
- Distinguish model recommendations from confirmation of harvesting safety.
- Identify and recover from input or connection errors.

These objectives assess both usability and appropriate interpretation of numerical estimates, image classifications, and decision-support labels.

6.3.2 Proposed Validation Procedure

Representative users could complete the growth prediction, disease detection, and harvest-readiness workflows during structured testing sessions. Evaluation should record task completion, input errors, assistance required, and completion time where relevant. Questionnaires or interviews could assess interface clarity, perceived usefulness, and trust in predictions.

Testing should include incomplete inputs and low-confidence disease predictions. Participants should also be informed that image classification is not laboratory confirmation and that a harvest-readiness prediction does not independently authorize robotic cutting.

6.3.3 Evidence and Limitations

The available evaluation material contains model-performance figures but no documented stakeholder survey, participant sample, usability results, or acceptance scores. Therefore, quantitative claims about user satisfaction or stakeholder acceptance cannot be justified.

A functioning interface demonstrates implementation, not successful stakeholder validation. A completed study should report participant characteristics, tasks, data-collection methods, findings, and limitations. If testing has not been conducted, it should be identified as future work.

6.3.4 Implications for Practical Use

Stakeholders must understand each output's meaning and limitations. Disease predictions should not be presented as confirmed diagnoses, growth estimates should be distinguished from derived indicators, and harvest-readiness labels should not imply guaranteed harvesting suitability or safety. These considerations help determine whether CinnaAI is appropriate for its intended agricultural workflow.

6.4 Integration Results

Integration results assess communication among the machine-learning functions, web application, supporting services, and hardware interface. Effective integration enables the individual components to operate within a shared monitoring and decision-support workflow.

6.4.1 Integration of Prediction Functions

CinnaAI provides three distinct outputs: estimated bark thickness, a predicted leaf-image category, and a harvest-readiness result. The shared frontend and backend support access to these functions while preserving their individual purposes.

These outputs should not be combined into a single decision without an explicitly defined and validated rule. Bark thickness, leaf condition, and harvest readiness represent different aspects of plant assessment and are not interchangeable.

6.4.2 Integration with Environmental Monitoring and Data Services

Growth prediction can use manually entered values or environmental readings from configured telemetry services. Successful integration requires valid, correctly mapped, and sufficiently current readings. Missing or outdated data should be identified rather than treated as reliable observations.

Authentication and storage services provide additional application support but require independent verification. A successful prediction does not guarantee that its associated record was saved, so inference and persistence outcomes should be reported separately.

6.4.3 Integration with Image Acquisition and Processing

The disease detection workflow connects image acquisition and processing with the retained classifier, whose input is a three-channel image of (224 \times 224) pixels.

The processing pipeline must produce the expected input representation while documenting the transformation from multispectral data to the final model input.

Reported checkpoint performance demonstrates results on the evaluated data but does not independently establish the contribution of multispectral imaging. This contribution requires a controlled comparison between suitable approaches using the same independent test cases.

6.4.4 Integration with Robotic Guidance

Growth estimates and harvest-readiness predictions can support decisions about potential harvesting actions, while the hardware interface provides a mechanism for dispatching supported controller commands.

However, generating a prediction, sending a command, and completing a physical harvesting operation are separate stages. Successful robotic integration requires verification of controller communication, actuator behaviour, positioning, cutting accuracy, and safety mechanisms. Unless supported by recorded physical trials, the implementation should be described as providing guidance and hardware-interface capabilities rather than validated autonomous harvesting.

7. DISCUSSION

7.1 Interpretation of Findings

This research developed a CinnaAI component integrating cinnamon growth prediction, disease detection, and harvest-readiness assessment with robotic guidance. The results demonstrate measurable model performance, although practical reliability depends on data quality, evaluation design, and field conditions.

The Random Forest growth model achieved an (R^2) of 0.8393, RMSE of 0.1648 mm, MAE of 0.1317 mm, and MSE of 0.0272 mm² across 2,000 evaluation records. These results indicate agreement with the evaluated bark-thickness targets. However, field accuracy depends on the targets' provenance and representativeness.

The EfficientNetB0 disease classifier achieved macro precision, recall, and F1-score of 84.07%, 86.11%, and 84.65% for the initial checkpoint. The exported model achieved 78.43%, 80.82%, and 78.60%, respectively, with a micro one-versus-rest ROC-AUC of 0.9824. The difference highlights the importance of evaluating the exact deployed artifact. Confidence and top two margin thresholds flag potentially uncertain predictions but do not guarantee correctness.

The harvest-readiness classifier achieved precision of 1.00, recall of 0.9231, F1-score of 0.9600, and ROC-AUC and average precision of 1.00 at a threshold of 0.50. At 0.80, precision remained 1.00, but recall decreased to 0.4615 and F1-score to 0.6316. This demonstrates the trade-off between stricter approval and missed positive cases. These results require further validation using representative field data before operational decisions are based on them.

7.2 Comparison with Literature Baselines and Existing Solutions

Conventional cinnamon monitoring may involve manual inspection, periodic measurements, and separate tools for individual tasks. Machine learning can standardize selected analytical processes and support consistent record-based monitoring, provided that models are developed and evaluated using suitable data.

Image-based disease classification offers a structured prediction for submitted images and may help identify cases requiring further inspection. However, it cannot replace expert assessment because disease symptoms can overlap with pest damage, environmental stress, and other conditions. Image quality, lighting, background, and differences between training and field images may also affect performance.

Comparisons with published disease-classification studies must consider class definitions, dataset size, class balance, preprocessing, data splitting, and metric calculation. Consequently, the exported model's macro F1-score of 78.60% and ROC-AUC of 0.9824 represent results from this project's evaluation, not proof of superiority over existing systems. The initial checkpoint's macro F1-score of 84.65% provides development evidence, but the deployed model is the appropriate reference for application performance.

Similarly, the growth model's (R^2) and error metrics indicate performance on the evaluated records but do not establish superiority over alternative regression methods without controlled comparisons on the same independent dataset. Its practical value depends on reliable bark-thickness measurements and generalization to different plants and growing conditions.

The harvest-readiness classifier introduces a learned decision-support approach alongside conventional rules and manual judgement. Its threshold results demonstrate how decision policies influence precision and recall. However, the appropriate threshold must be determined using reliable readiness labels, real harvesting requirements, and the consequences of incorrect recommendations.

Overall, the component's main potential advantage is its integration of complementary monitoring functions within one software workflow. Comparative superiority remains unproven until controlled baseline comparisons and representative field evaluations are completed.

7.3 Research Contribution

The principal research contribution is the implementation and evaluation of a machine learning-based approach that combines cinnamon growth estimation, disease classification, and harvest-readiness assessment.

First, the Random Forest regressor estimates bark thickness from selected plant and environmental features, providing a basis for further growth-monitoring research. Its generalizability requires validation across plants, locations, seasons, and measurement conditions.

Second, the disease-classification pipeline identifies seven cinnamon-related categories, including healthy cinnamon, selected disease or pest-related conditions, and non-cinnamon images. Confidence and margin checks flag potentially uncertain predictions, although their effectiveness requires separate evaluation.

Third, harvest readiness is treated as a distinct classification task rather than being inferred directly from bark thickness. The results at thresholds of 0.50 and 0.80 demonstrate how decision thresholds affect precision and recall.

These contributions represent applied research and system development rather than the introduction of a new machine-learning algorithm or proof of universal generalizability. The research value lies in combining task-specific models, application integration, and an evaluation framework while identifying requirements for further validation.

7.4 Product Contribution

The product contribution is a software component that connects predictive models with a shared workflow for cinnamon monitoring and harvest decision support.

The disease detection function classifies uploaded images into seven categories and flags predictions meeting configured uncertainty conditions. Its output can support further inspection but should not be treated as a definitive diagnosis.

The growth function estimates bark thickness and may derive a growth indicator from that estimate. The indicator's formula and interpretation should be documented to distinguish it from a direct measurement or a probability of harvest readiness.

The harvest-readiness function produces a separate classification based on a configurable decision threshold. A stricter threshold may reduce incorrect positive recommendations but can also increase missed positive cases. The interface should explain this trade-off without implying that a prediction guarantees harvesting suitability or safety.

Together, these functions provide an application-oriented decision-support workflow. Their practical benefit remains dependent on model reliability, data quality, and validation with intended users.

7.5 Limitations and Threats to Validity

Several limitations affect the interpretation of the results and the component's readiness for operational use.

Data representativeness. Plant age, growing conditions, disease presentation, lighting, leaf orientation, and background can vary substantially. The reported aggregate metrics do not establish that all relevant field conditions are represented. Data collection, labelling, sample distribution, and measurement procedures should be documented.

Growth target provenance. The growth evaluation reports 2,000 records, but the supplied figure includes the labels “Generated target bark thickness (mm)” and “saved synthetic audit.” These descriptions must be reconciled with the claim that real data were used. If targets are measured values, their collection procedure should be documented; if any are generated, their role must be stated explicitly. Until clarified, the reported metrics should not be interpreted as definitive evidence of field accuracy.

Dataset splitting and leakage. If multiple observations from the same plant, location, or acquisition session appear in both training and evaluation sets, performance may be overestimated. The available summary does not establish whether grouped splitting was used. Future evaluation should consider plant-, location-, or session-level separation and an independent test set.

Disease classification limitations. The classifier predicts image-level categories but does not segment lesions or quantify disease severity. Similar symptoms may have different causes. The lower metrics of the exported model compared with the initial checkpoint also warrant investigation. Further evaluation should include per-class metrics, confusion matrices, failure cases, and images collected under varied field conditions.

Confidence and threshold limitations. Confidence and margin checks flag predictions based on fixed rules but do not guarantee that accepted predictions are correct. Likewise, harvest-readiness thresholds cannot compensate for unreliable labels or unrepresentative data. The perfect precision and ROC-AUC reported for harvest readiness must be interpreted alongside test-set size, class distribution, label quality, and test independence.

Agronomic validity. Reproducing evaluation labels does not necessarily establish true harvesting readiness. The criteria used to define readiness, the reliability of those labels, and the consequences of incorrect decisions require further investigation and field validation.

Stakeholder and field evidence. The available results describe model performance and implementation but do not establish improved agricultural outcomes or documented user acceptance. Structured user studies and field trials are needed to assess usability, trust, decision quality, and practical benefits.

These limitations define the scope of the current findings and identify the additional evidence required to establish generalizability and operational suitability.

7.6 Implications

The component illustrates how image classification and numerical prediction can support cinnamon monitoring through a shared digital workflow. Disease predictions may identify images requiring inspection, growth estimates may provide quantitative monitoring references, and harvest-readiness outputs may contribute to decision-making.

For growers and agricultural practitioners, these outputs should complement rather than replace expert judgement. Disease classifications require confirmation where appropriate, bark-thickness estimates should be considered alongside reliable measurements, and harvest recommendations should be assessed against established agronomic criteria.

In conclusion, the research establishes an application-oriented foundation for integrating cinnamon growth estimation, disease classification, and harvest-readiness assessment. The reported metrics provide initial evidence of model performance, while differences between model versions and unresolved questions about data provenance require cautious interpretation. Independent evaluation, field testing, stakeholder validation, and physical robotic trials remain necessary to establish generalizability and practical impact.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The main aim of this research was to develop a machine learning-based component to support cinnamon plant monitoring through growth prediction, disease detection, and harvest guidance. The component was designed to bring these functions together within a common software workflow, using image-based analysis, predictive modelling, and decision-support mechanisms to assist the interpretation of cinnamon plant conditions.

The growth prediction function used a Random Forest regression model to estimate cinnamon bark thickness. The reported evaluation results showed an \(R^2\) value of 0.8393, an MAE of 0.1317 mm, an RMSE of 0.1648 mm, and an MSE of 0.0272 mm² across 2,000 paired evaluation records. These results indicate that the model captured a substantial proportion of the variation in the evaluated target values. The error metrics also provide a quantitative basis for assessing the differences between predicted and target bark thickness. However, the applicability of these results to real plantation conditions depends on confirming the provenance of the target measurements and validating the model on independent field observations.

The disease detection function used an EfficientNetB0-based image classification approach to distinguish seven cinnamon-related image categories, including healthy plants, selected disease or pest-related conditions, and non-cinnamon images. The exported final model achieved a macro precision of 78.43%, macro recall of 80.82%, macro F1-score of 78.60%, and micro one-versus-rest ROC-AUC of 0.9824. Although the ROC-AUC indicates strong aggregate discrimination on the evaluated data, the classification results also show the importance of examining performance across individual classes. The initial checkpoint achieved a higher macro F1-score of 84.65%, making it important to evaluate the exact model artifact used in the application and investigate the differences between the two versions.

The harvest readiness function was developed to classify whether a plant met the model's harvest readiness criteria. At a decision threshold of 0.50, the reported precision was 1.00, recall was 0.9231, and F1-score was 0.9600. At the stricter threshold of 0.80, precision remained at 1.00, while recall decreased to 0.4615. This demonstrates how the decision threshold influences the balance between identifying harvest-ready plants and avoiding incorrect positive recommendations. These results are promising within the evaluated dataset but require additional validation against reliable agronomic readiness criteria before the model can be relied upon for real harvesting decisions.

Overall, the research demonstrates the implementation and evaluation of complementary machine learning functions within a cinnamon monitoring component. Its principal outcome is an integrated software-based approach that combines numerical growth estimation, image classification, and harvest readiness decision support. The results provide a foundation for further development, while data verification, independent testing, multispectral evaluation, and field validation remain necessary to establish broader reliability and practical agricultural value.

8.2 Achievement of objectives

The achievement of the research objectives is assessed by considering the implemented functionality, available evaluation evidence, and the remaining validation requirements. The objectives were directed towards growth estimation, disease detection, harvest readiness assessment, and integration of the resulting functions into the wider CinnaAI application.

Objective 1: Develop a machine learning-based cinnamon growth prediction function.

The growth estimation function was implemented using a Random Forest regression model. It accepts plant and environmental inputs and returns an estimated bark thickness, together with a derived growth indicator and status. The reported audit metrics provide an initial assessment of the model's behaviour. However, the project report identifies the growth targets as synthetically generated rather than independently measured biological observations. Consequently, the objective of implementing a growth estimation function was achieved at the software level, while reliable real-world growth prediction remains to be established through validation against actual bark thickness measurements.

Objective 2: Develop a machine learning-based cinnamon disease detection function.

The disease detection pipeline was implemented using EfficientNetB0 for seven-class image classification. The application prepares uploaded images, generates class predictions, and applies confidence and margin checks to flag uncertain results. The implementation therefore establishes the core image classification workflow. However, the original training image dataset and complete final evaluation outputs were not available in the inspected project files. The reported classification metrics should be retained only if their supporting evaluation records can be verified. The objective of implementing disease classification was achieved, while the model's real-world diagnostic reliability requires further validation.

Objective 3: Develop a harvest readiness prediction and guidance function.

A Random Forest classifier was implemented to estimate a peeling-readiness label from four input features: age, rainfall index, phenological stage, and bark browning percentage. The predicted class probability is converted into a readiness status and action recommendation. However, the project report identifies the training data and labels as synthetic, with labels constructed from predefined threshold rules rather than independently recorded physical peeling tests. Therefore, the objective of implementing harvest readiness decision support was achieved at the software level, but its agronomic accuracy and usefulness for actual harvesting remain unverified.

Objective 4: Incorporate multispectral acquisition and processing into the monitoring workflow.

Multispectral acquisition and processing are part of the intended monitoring workflow. Their purpose is to provide image information for plant condition analysis and support further development of machine learning-based monitoring. However, the contribution of the spectral information to the final disease classification performance must be documented and evaluated separately. In particular, the implemented disease model's three-channel RGB input should not be described as direct processing of all multispectral bands without evidence of that connection.

Objective 5: Integrate the component with the wider CinnaAI application.

The implemented component connects model-related functions with a web application and backend services. The growth and disease workflows provide structured outputs, while the harvest workflow provides readiness and action labels. Authentication, prediction history, data persistence, and supported serial communication are incorporated into the wider application architecture. Nevertheless, the project report identifies limitations in harvest CSV persistence and the integration of the separate robotic service. The achievement is therefore best described as an application-level integration prototype rather than a fully validated autonomous harvesting solution.

Overall, the objectives concerning the implementation of machine learning functions and their integration into a common software workflow have been substantially addressed. The objectives relating to verified real-world predictive performance, demonstrated multispectral improvement, and dependable physical robotic harvesting require additional evidence.

8.3 Individual research and product contributions

The individual contribution of this research is the development of a cinnamon monitoring component that brings together growth estimation, image-based disease classification, and harvest readiness decision support within the CinnaAI system.

Research contributions

The principal research contributions are as follows:

1. Growth estimation: Implementation of a Random Forest regression pipeline to estimate bark thickness from the available input features and produce a derived growth indicator.

2. Disease classification: Implementation of an EfficientNetB0-based image classification workflow for seven cinnamon-related categories, including healthy, disease or pest-related, and non-cinnamon classes.

3. Prediction uncertainty handling: Introduction of confidence and top-two probability margin checks to flag disease predictions that may require further review.

4. Harvest readiness assessment: Implementation of a separate classification model that estimates peeling readiness from selected plant and environmental indicators and converts its output into a decision-support recommendation.

5. Multispectral processing workflow: Incorporation of multispectral acquisition and processing into the broader monitoring design, creating a basis for future investigation of spectral features for cinnamon health assessment.

These contributions represent applied machine learning and system development. They do not establish a new machine learning algorithm or demonstrate that the models outperform existing research methods. The scientific strength of the contribution will depend on completing independent evaluation with verified measurements and appropriately labelled images.

Product contributions

The main product contributions are:

- A common software workflow for accessing cinnamon growth, disease, and harvest-related predictions.

- A disease image analysis interface that returns structured classifications and uncertainty-related information.
- A growth estimation interface that presents numerical outputs and associated growth indicators.
- A harvest readiness interface that translates model outputs into status and action recommendations.
- Integration with the wider application's frontend, backend, authentication, and data services.
- A foundation for extending harvest guidance towards robotic systems through supported hardware communication interfaces.

8.4 Future directions

Although the component provides a foundation for machine learning-based cinnamon monitoring, further work is required to improve predictive reliability, validate the use of multispectral information, and establish the practical value of harvest guidance.

1. Validate growth prediction using real measurements

The first priority is to develop a dataset containing independently measured bark thickness and corresponding plant observations. Measurements should be recorded using a consistent procedure, with information about plant identity, location, observation date, and relevant environmental conditions. The model should then be evaluated on independent plants or locations to assess whether it generalises beyond its development data.

Future experiments should compare Random Forest regression against appropriate baselines, including a simple linear regression model and a direct prediction rule where applicable. Performance should be reported using \(R^2\), MAE, and RMSE, with uncertainty estimates where possible. Synthetic-data results should be clearly separated from results obtained using real measurements.

2. Improve disease classification and field validation

The disease classification model should be evaluated using a verified image dataset with reliable labels and documented class distributions. Images should cover variation in illumination, leaf orientation, background, disease severity, plant age, and acquisition conditions. Where feasible, labels should be checked by a suitably qualified agricultural expert.

The next evaluation should report per-class precision, recall, and F1-score, alongside confusion matrices and representative misclassified examples. The final exported

model should be tested using exactly the same preprocessing pipeline employed by the application. The difference between the initial checkpoint and exported model should also be investigated.

Additional work should examine confidence calibration and the effectiveness of the low-confidence and probability-margin rules. An explicit unknown or uncertain outcome may help reduce the risk of presenting unreliable classifications as definitive diagnoses.

3. Establish the contribution of multispectral imaging

Further experiments should document the spectral acquisition hardware, available bands, image calibration procedure, preprocessing operations, and the relationship between acquired data and model inputs. If multispectral features are intended to support disease detection, they should be evaluated directly rather than assuming that their presence in the overall workflow improves classification.

A controlled comparison between RGB-based classification and multispectral-based classification would help determine whether the additional spectral information improves disease recognition. Both approaches should be evaluated on the same independent test data, using the same class definitions and metrics. Future research could also investigate whether selected spectral indices or band combinations provide useful information for identifying early plant stress before symptoms are clearly visible.

4. Validate harvest readiness against agronomic criteria

The harvest readiness classifier should be developed and tested using independently recorded readiness labels. Actual observations should be collected from cinnamon plants at different stages, and the criteria for determining peeling readiness should be documented with appropriate agricultural expertise.

The model should then be evaluated using a separate test set, with particular attention to false-positive and false-negative decisions. The thresholds used to convert prediction scores into readiness recommendations should be selected according to the consequences of incorrect decisions rather than model scores alone. The derived recommendation should be compared with established harvesting practice before it is used to guide real operations.

5. Extend and validate robotic harvest guidance

Future development should connect validated readiness decisions with a clearly defined robotic workflow. This would require reliable plant localisation, stem geometry estimation, positioning, motion planning, and controlled interaction with the plant. Any cutting or peeling mechanism would need separate mechanical and safety validation.

Initial trials should be conducted in controlled conditions before progressing to plantation environments. Relevant measures could include target localisation error, successful positioning rate, harvesting cycle time, plant damage, command response, and safe stopping behaviour. The software should require appropriate safeguards before any physical harvesting action is authorised.

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GLOSSARY

Term	Definition
Artificial Intelligence (AI)	A field of computing that enables machines to perform tasks that typically require human intelligence.
Machine Learning (ML)	A branch of AI that enables computer systems to learn patterns from data and make predictions or classifications.
Deep Learning (DL)	A type of machine learning that uses multi-layered neural networks to learn complex patterns from data.
Cinnamon	A plant belonging to the genus Cinnamomum, cultivated for its aromatic bark and other useful products.
Growth Prediction	The process of estimating future plant growth using collected data and machine learning models.
Disease Detection	The identification of possible plant diseases through image analysis and classification techniques.
Robotic Harvest Guidance	The use of computational predictions and sensor information to assist in determining suitable harvesting conditions and actions.
Multispectral Imaging	An imaging technique that captures information across multiple wavelength bands of light, including wavelengths beyond visible light.
Image Processing	The manipulation and analysis of digital images to extract useful information.
Computer Vision	A field of AI that enables computers to interpret and analyze images and videos.
Convolutional Neural Network (CNN)	A deep learning model commonly used to identify patterns and features in images.
EfficientNetB0	A convolutional neural network architecture designed to achieve a balance between model accuracy and computational efficiency.
Random Forest (RF)	A machine learning algorithm that combines multiple decision trees to make predictions or classifications.
Feature Extraction	The process of identifying and converting relevant information from raw data into useful model inputs.
Dataset	A collection of data used to train, validate, or evaluate a machine learning model.
Training Data	Data used to teach a machine learning model to recognize patterns.